2.3 Stabilizing the quiescent working point Q

稳定放大电路工作点Q

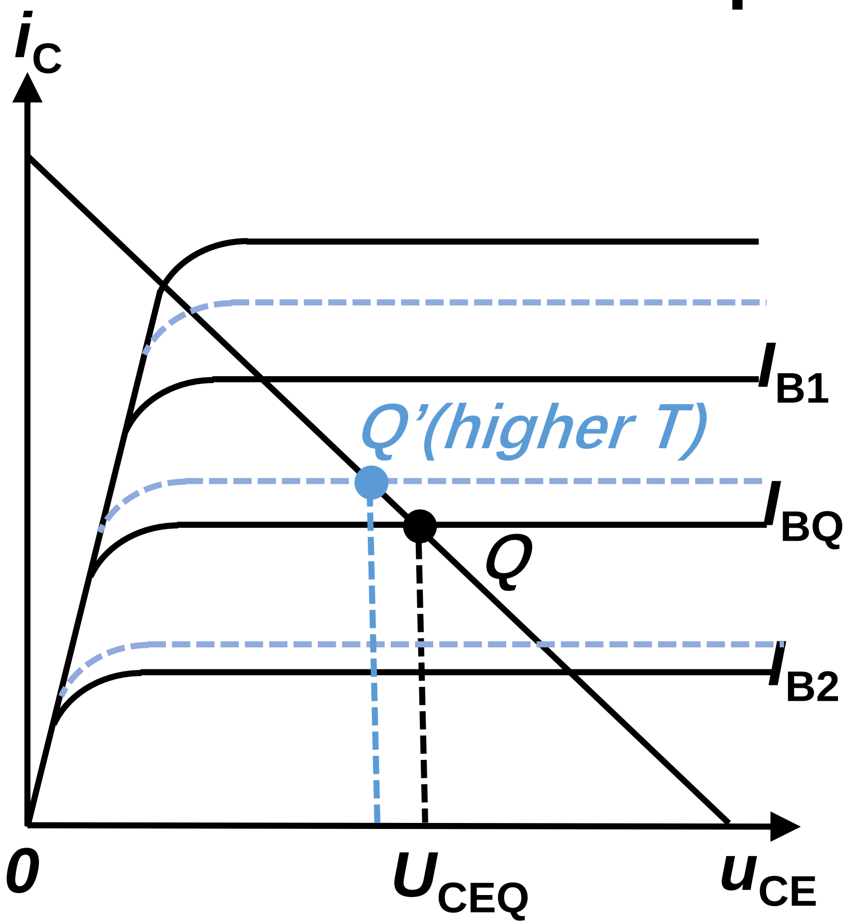
Many factors can affect the quiescent working point Q:

1. Aging 老化

2. Fluctuation of supply voltage V_{CC}

3. Temperature effect

Temperature effect



Temperature ↑:

For the same I_{BQ} value

$I_{CQ}\uparrow$ $U_{CEQ}\downarrow$ $\beta\uparrow$

Amplifier circuit is unstable

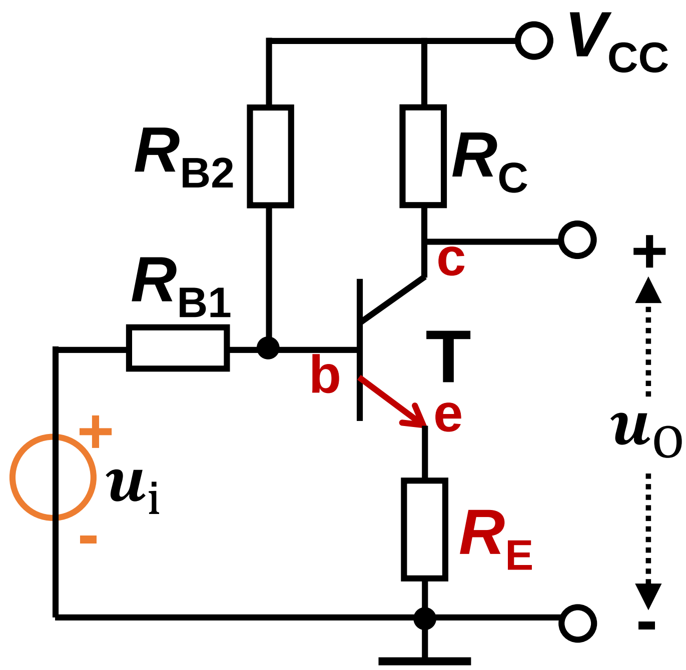
Quiescent working points change!

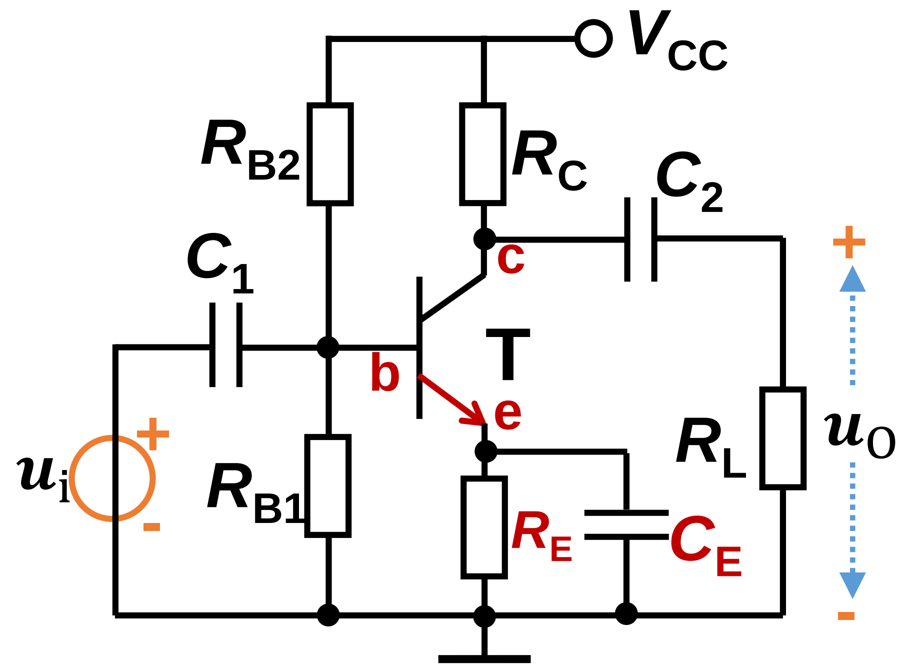
AC voltage gain changes!

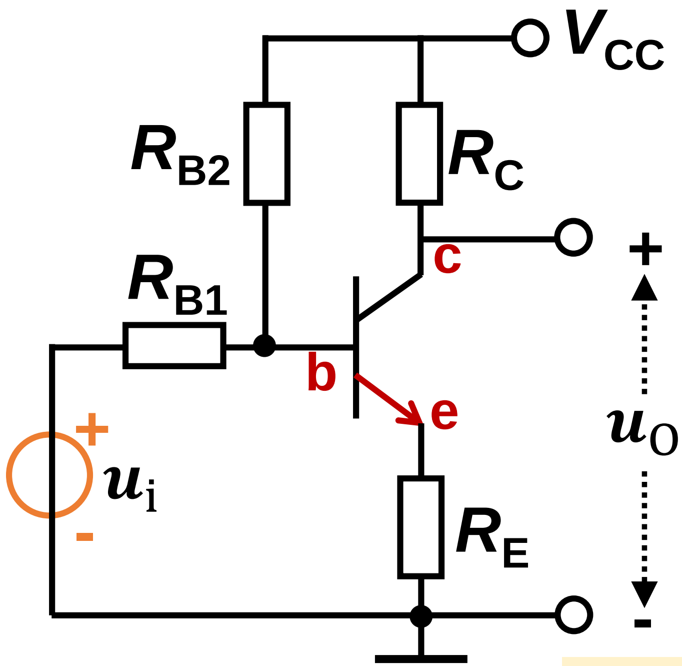
2.3.1 Negative feedback to stabilize the quiescent working point Q

Direct coupling

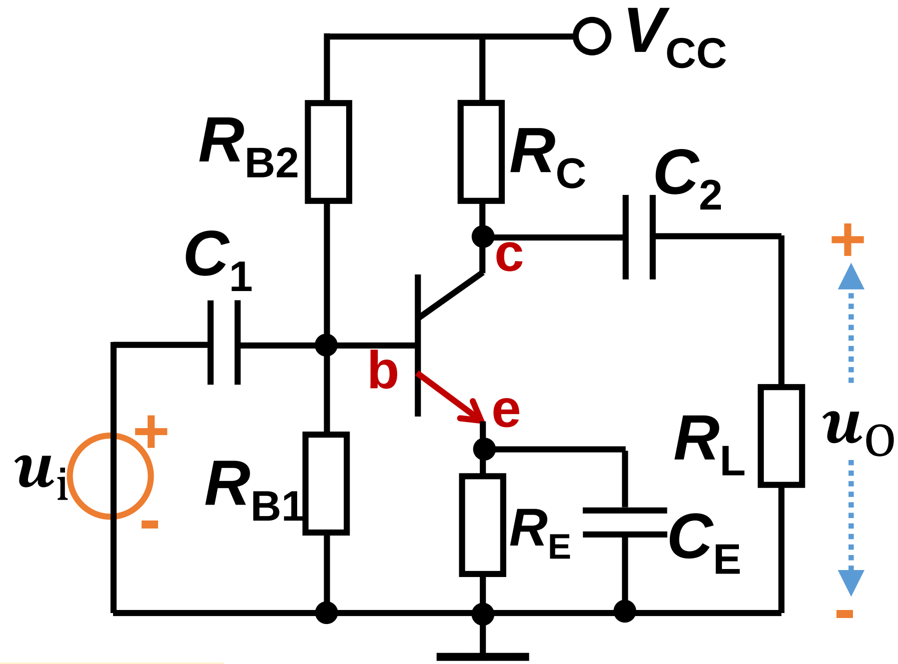
Resistance-capacitance coupling





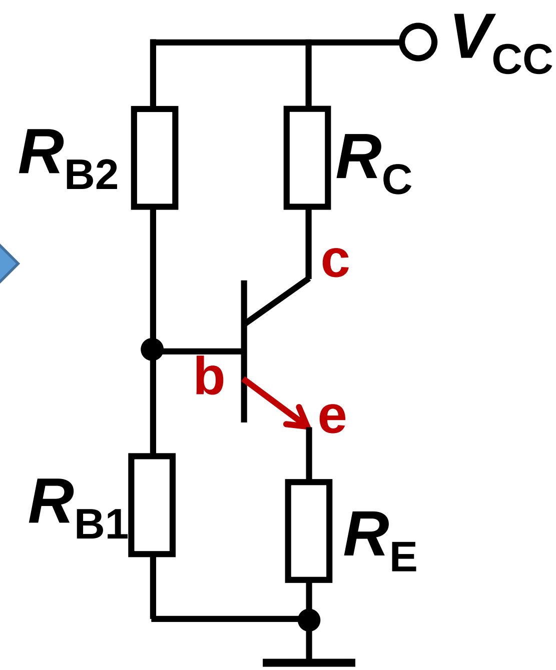


V_{CC}
R_{B2}
R_C
R_{B1}
c
b
e
R_E
+
u_i
-
+
u_O
-



V_{CC}
R_{B2}
R_C
C_2
C_1
c
b
e
+
u_i
-
R_{B1}
R_E
C_E
R_L
+
u_O
-

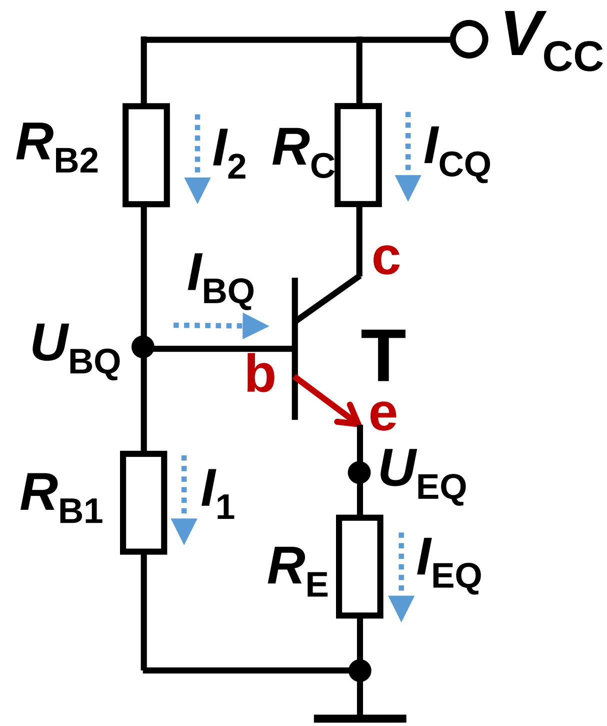
DC circuit



V_{CC}
R_{B2}
R_C
c
b
e
R_{B1}
R_E

Working principle

$$I_2 = I_1 + I_{BQ}$$



Because I_{BQ} is very small

I_2 **and** $I_1 \gg I_{BQ}$

$I_2 \approx I_1$ (Approximation)

$$U_{BQ} \approx \frac{R_{B1}}{R_{B1} + R_{B2}} \cdot V_{CC}$$

Then U_{BQ} is unrelated to temperature

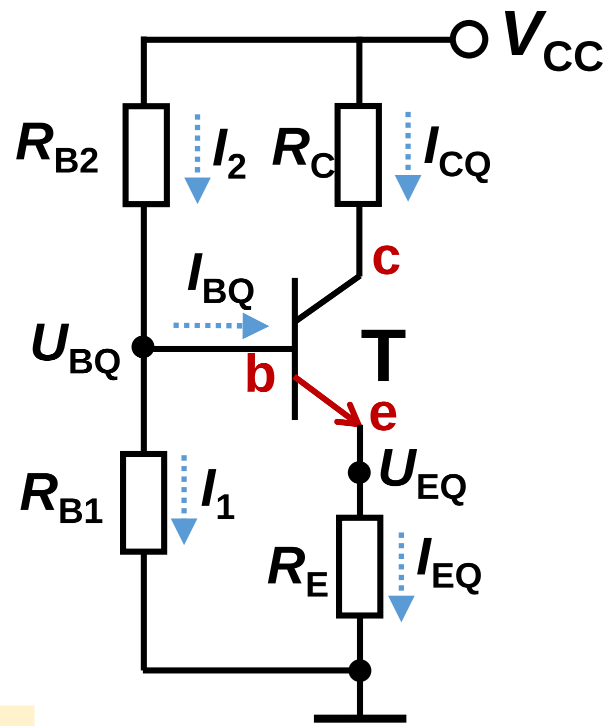
$T \uparrow \Rightarrow I_{CQ} \uparrow \Rightarrow I_{EQ} \uparrow \Rightarrow U_{EQ} \uparrow$

$\Downarrow$ $U_{BEQ} = U_{BQ} - U_{EQ}$

$I_{CQ} \downarrow \Leftarrow I_{BQ} \downarrow \Leftarrow U_{BEQ} \downarrow$

Feedback 反馈

R_E: Transport output signal (I_C) back to the input circuit, and in turn affect the input signal (U_{BEQ})



Negative Feedback 负反馈

The feedback make the variation of output signal smaller

DC Negative Feedback 直流负反馈

$T\uparrow \Rightarrow I_{CQ}\uparrow \Rightarrow I_{EQ}\uparrow \Rightarrow U_{EQ}\uparrow$

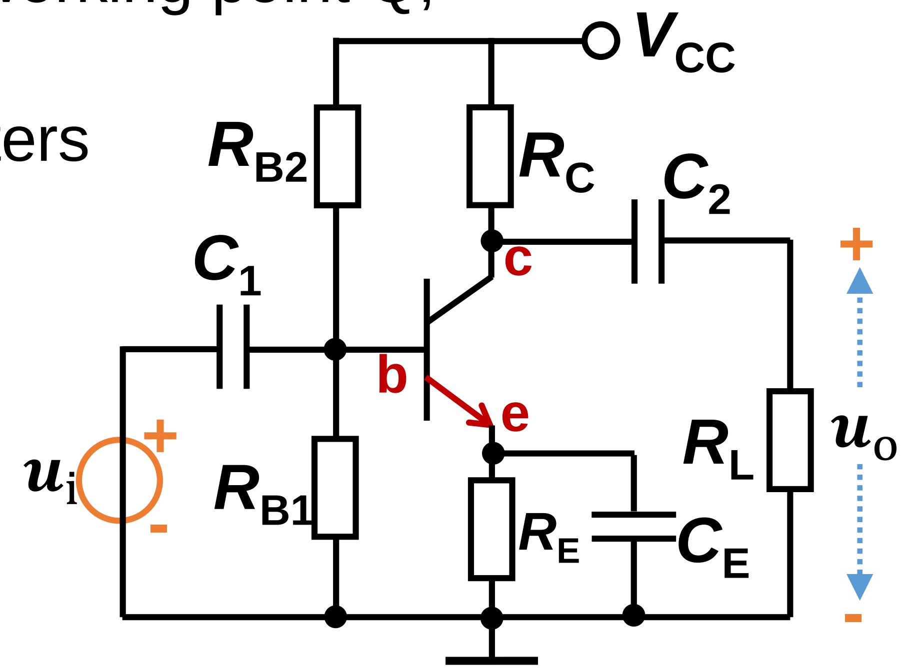
$\Downarrow$ $U_{BEQ} = U_{BQ} - U_{EQ}$

$I_{CQ}\downarrow \Leftarrow I_{BQ}\downarrow \Leftarrow U_{BEQ}\downarrow$

[Q1] As shown in the figure, R_{B2}=39kΩ, R_{B1}=10kΩ, R_C=2.7kΩ, R_E=1kΩ, R_L=5.1kΩ and V_{CC}=15V, transistor has β =100, $r_{bb'}$=300Ω and U_{BEQ}=0.7V, assume capacitors C_1 and C_2 have no resistance to AC signals, ask:

(1) Parameters of quiescent working point Q;

(2) Dynamic working parameters A_u, R_i, R_o.



(1) Parameters of static working point Q

1st approach: estimation method

$$I_2, I_1 \gg I_{BQ} \Rightarrow I_2 \approx I_1$$

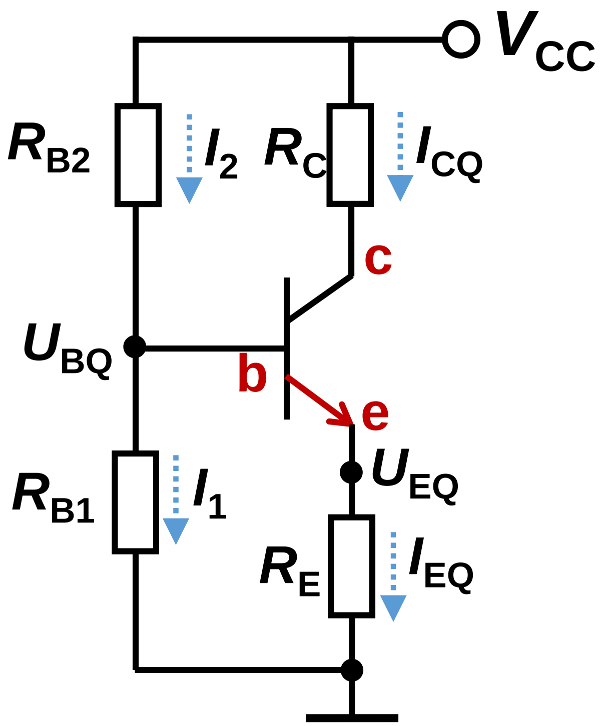
$$U_{BQ} \approx \frac{R_{B1}}{R_{B1}+R_{B2}} \cdot V_{CC} = 3.06\text{V}$$

$$I_{EQ} = \frac{U_{EQ}}{R_E} = \frac{U_{BQ} - U_{BEQ}}{R_E} = 2.36\text{mA}$$

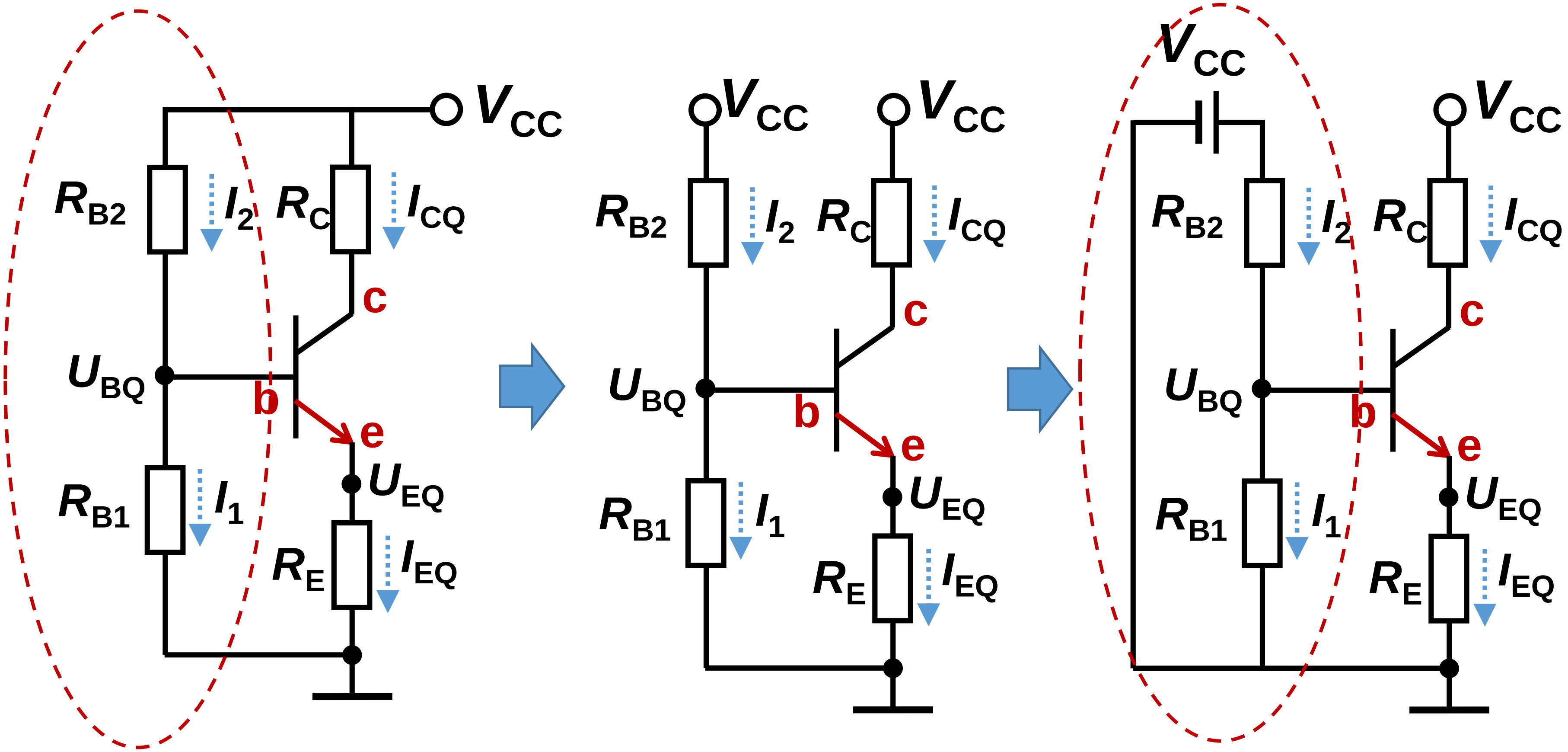
$$I_{BQ} = \frac{I_{EQ}}{\beta + 1} = 0.02\text{mA}$$

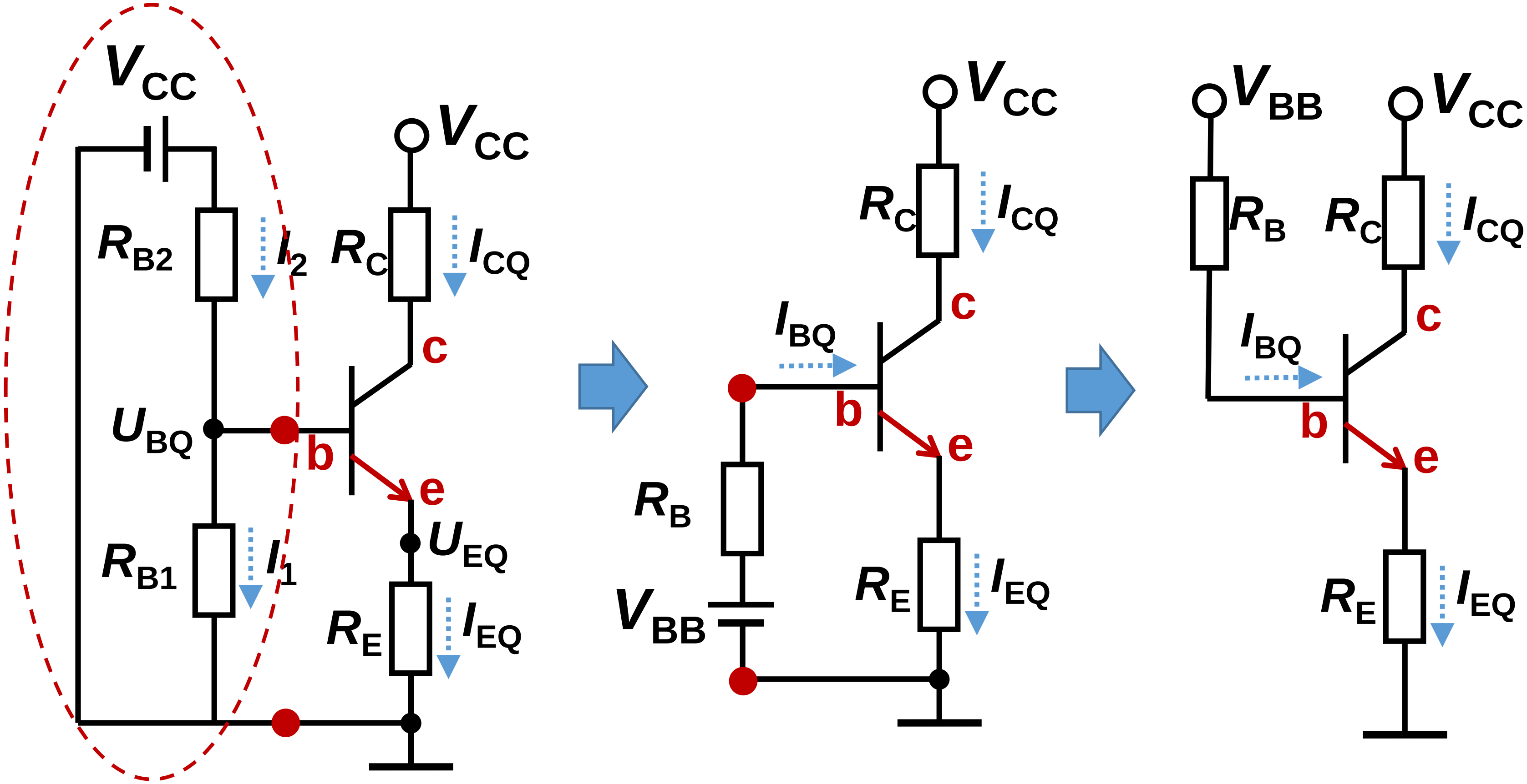
$$I_{CQ} = \beta \frac{I_{EQ}}{\beta + 1} = 2.34\text{mA}$$

$$U_{CEQ} = V_{CC} - I_{CQ}R_C - I_{EQ}R_E = 6.32\text{V}$$

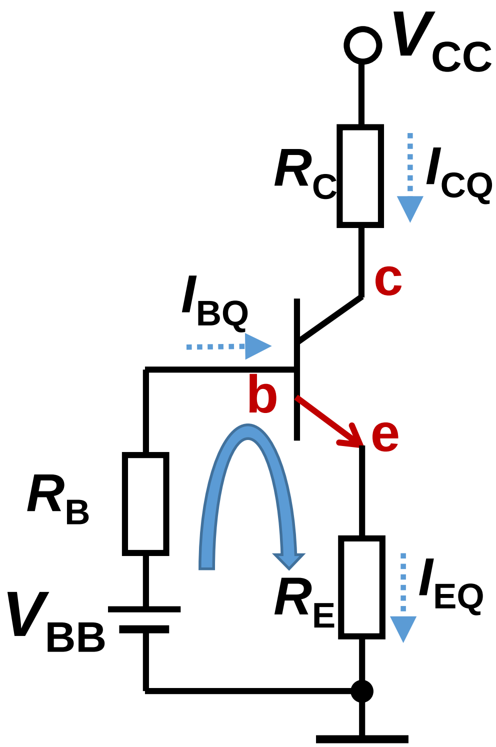


2^{nd} approach: Thevenin's theorem 戴维南定理





$$V_{\mathrm{BB}} = \frac{R_{\mathrm{B1}}}{R_{\mathrm{B1}} + R_{\mathrm{B2}}} \cdot V_{\mathrm{CC}} = 3.06\mathrm{V} \qquad R_{\mathrm{B}} = R_{\mathrm{B1}} || R_{\mathrm{B2}} = 7.96\mathrm{k}\Omega$$



$$V_{\mathrm{BB}} = \frac{R_{\mathrm{B1}}}{R_{\mathrm{B1}} + R_{\mathrm{B2}}} \cdot V_{\mathrm{CC}} = 3.06\mathrm{V}$$

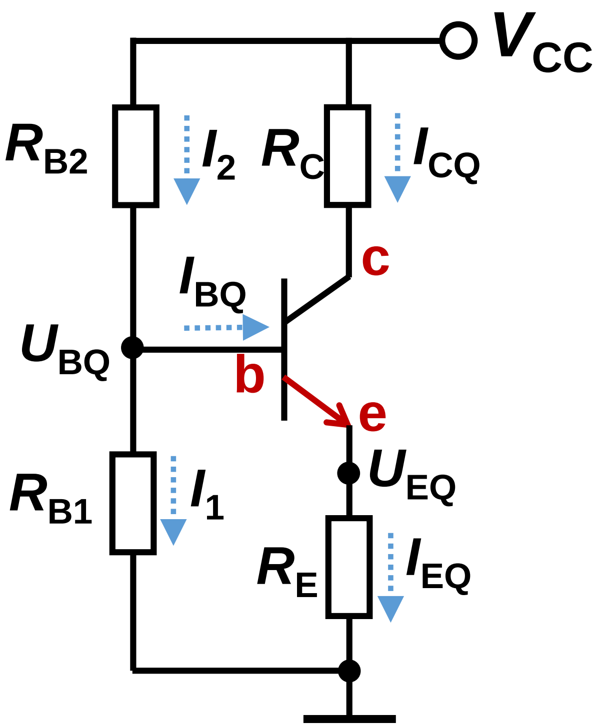
$$R_{\mathrm{B}} = R_{\mathrm{B1}} || R_{\mathrm{B2}} = 7.96\mathrm{k}\Omega$$

$$\begin{cases} V_{\mathrm{BB}} - I_{\mathrm{BQ}} R_{\mathrm{B}} - U_{\mathrm{BEQ}} - I_{\mathrm{EQ}} R_{\mathrm{E}} = 0 \\ I_{\mathrm{EQ}} = (1+\beta) I_{\mathrm{BQ}} \end{cases}$$

$$\Downarrow$$

$$I_{\mathrm{BQ}} = \frac{V_{\mathrm{BB}} - U_{\mathrm{BEQ}}}{R_{\mathrm{B}} + (1+\beta) R_{\mathrm{E}}}$$

The 3rd approach: A violent approach暴力求解法

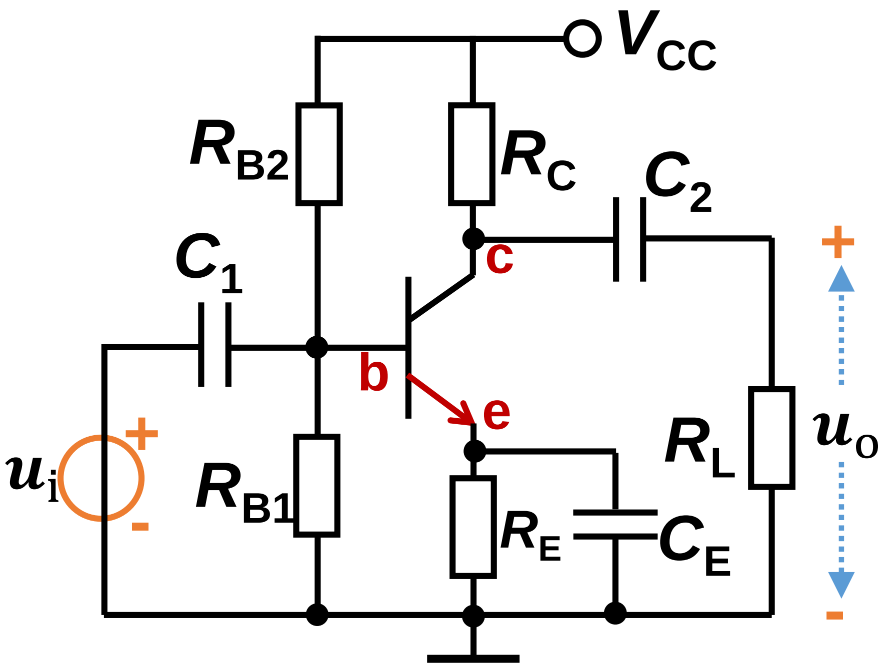


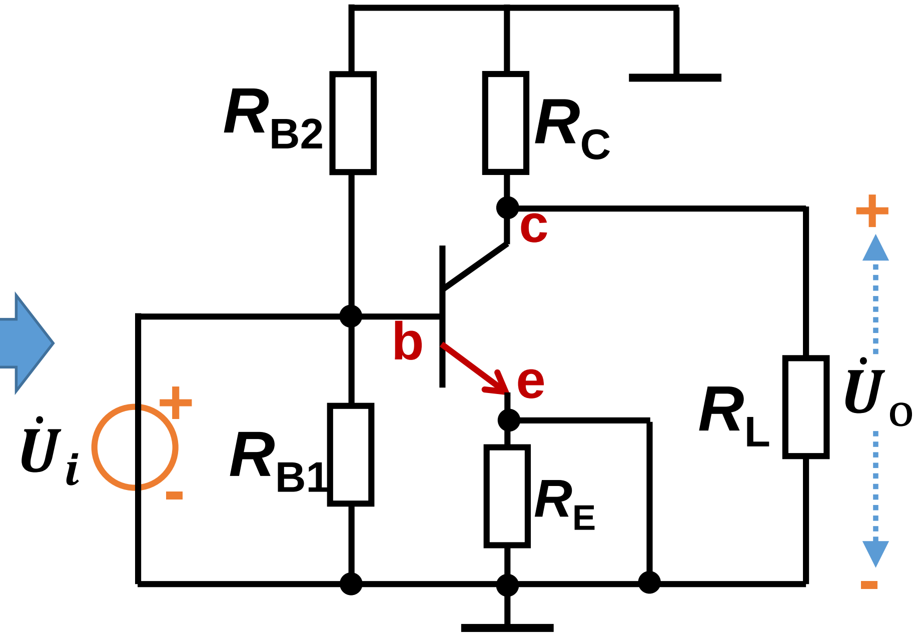
$$\begin{cases} V_{\mathrm{CC}} - I_2 R_{\mathrm{B2}} - I_1 R_{\mathrm{B1}} = 0 \\ U_{\mathrm{BEQ}} + I_{\mathrm{EQ}} R_{\mathrm{E}} - I_1 R_{\mathrm{B1}} = 0 \\ I_2 = I_1 + I_{\mathrm{BQ}} \\ I_{\mathrm{EQ}} = (1+\beta) I_{\mathrm{BQ}} \end{cases}$$

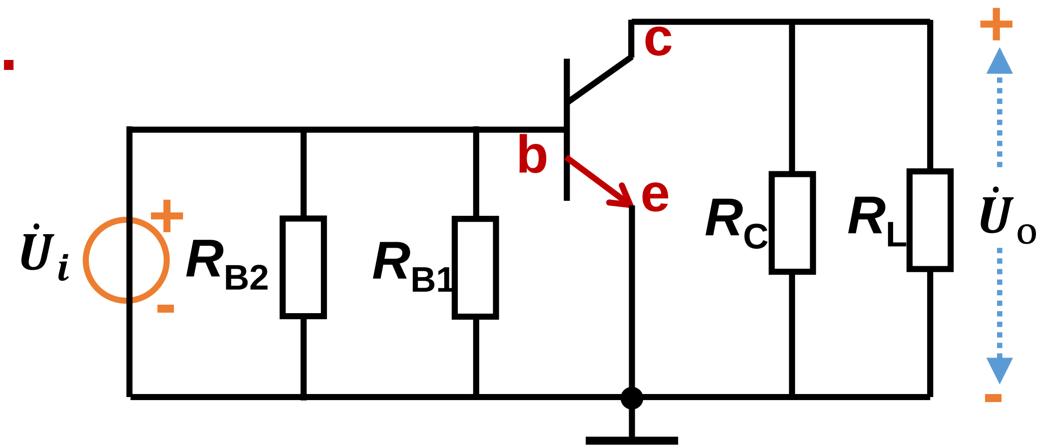
$$\Downarrow$$

$$I_{\mathrm{BQ}} = \frac{\dfrac{R_{\mathrm{B1}}}{R_{\mathrm{B1}} + R_{\mathrm{B2}}} \cdot V_{\mathrm{CC}} - U_{\mathrm{BEQ}}}{R_{\mathrm{B1}} || R_{\mathrm{B2}} + (1+\beta) R_{\mathrm{E}}}$$

(2) A_u, R_i, R_o

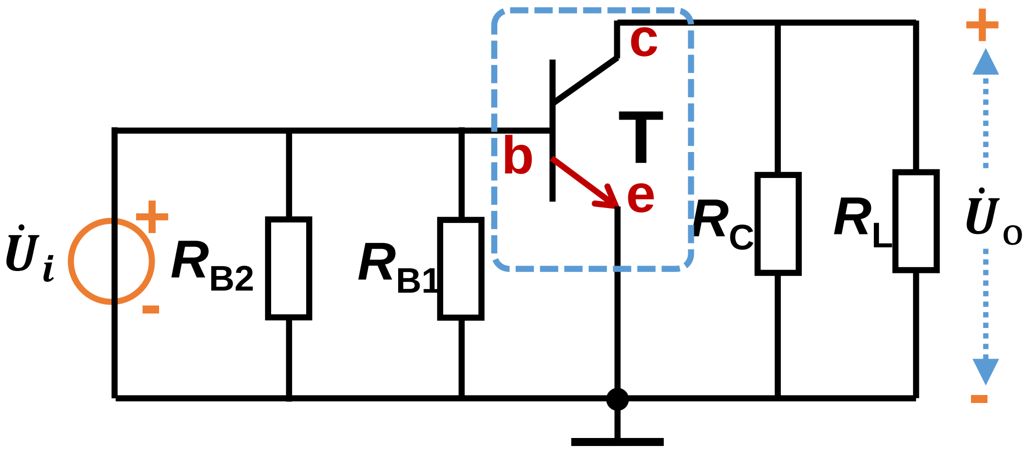


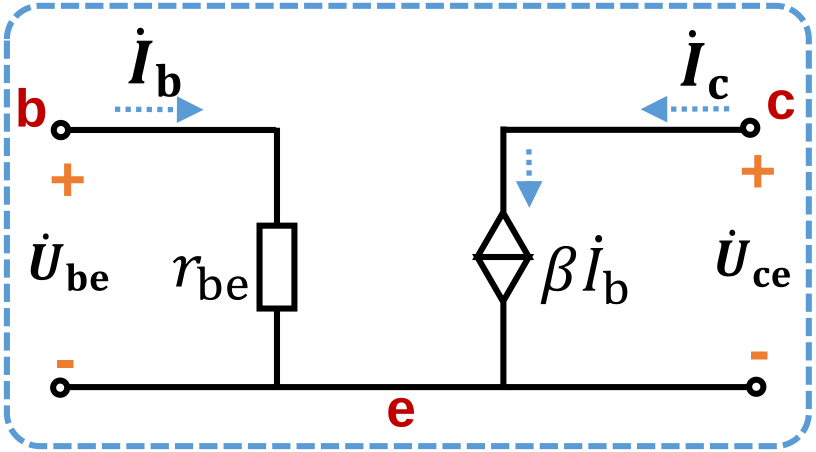


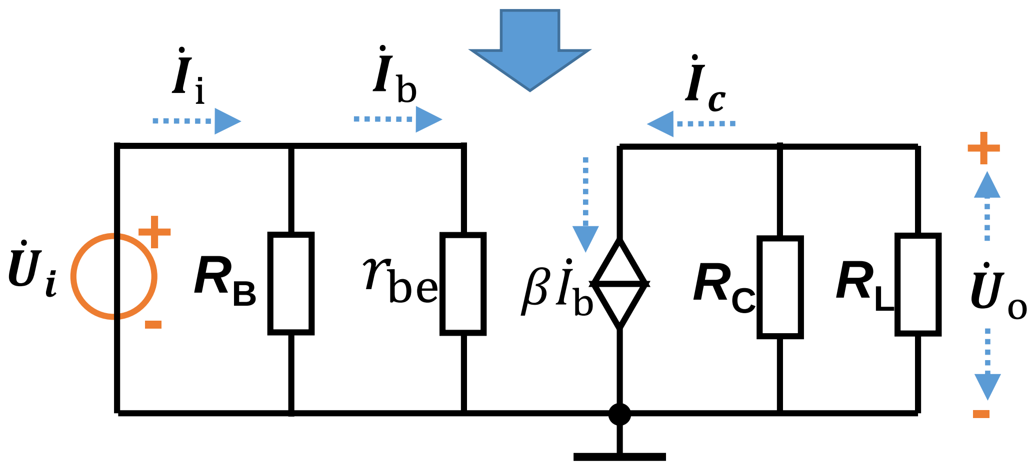


1st step: get AC circuit.

2nd step: get h-model.





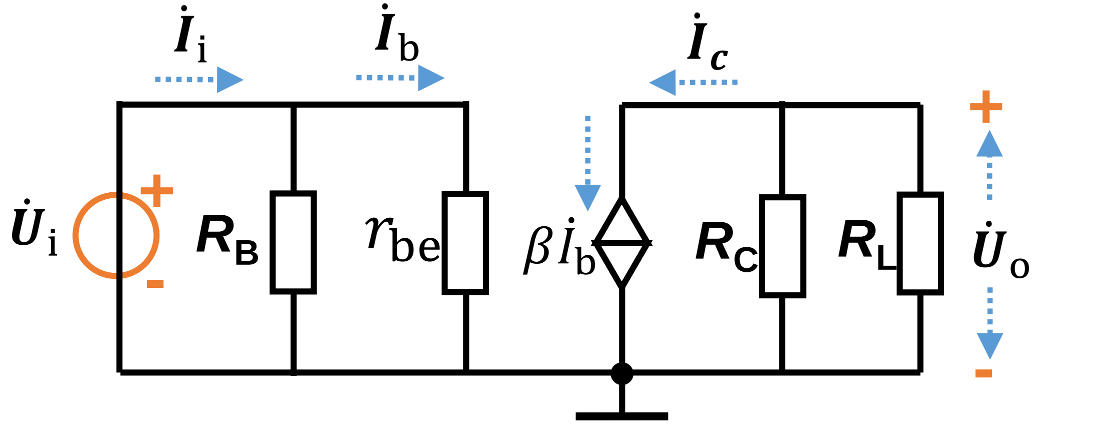


$$R_B = R_{B1} || R_{B2} = 7.96\text{k}\Omega$$

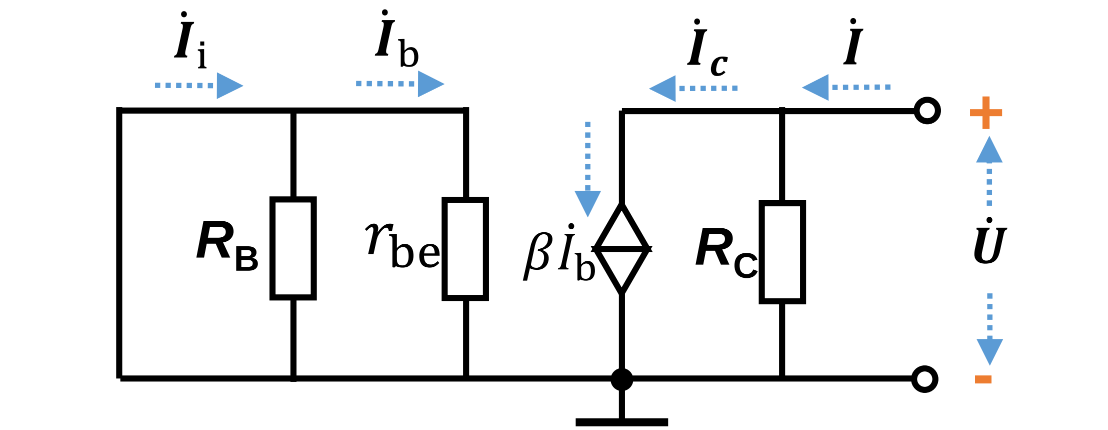
$$r_{be} = r_{bb'} + \beta \frac{U_T}{|I_{CQ}|}$$

$$= 300 + 100 \frac{0.026}{0.00234} \approx 1.4\ \text{k}\Omega$$

Voltage gain: $$\dot{A}_u = -\frac{\beta R_L'}{r_{be}} = -\frac{\beta (R_L || R_C)}{r_{be}} \approx -126$$



Input resistance: $R_{\rm i} = R_{\rm B}||r_{\rm be} = 1.2\ {\rm k}\Omega$

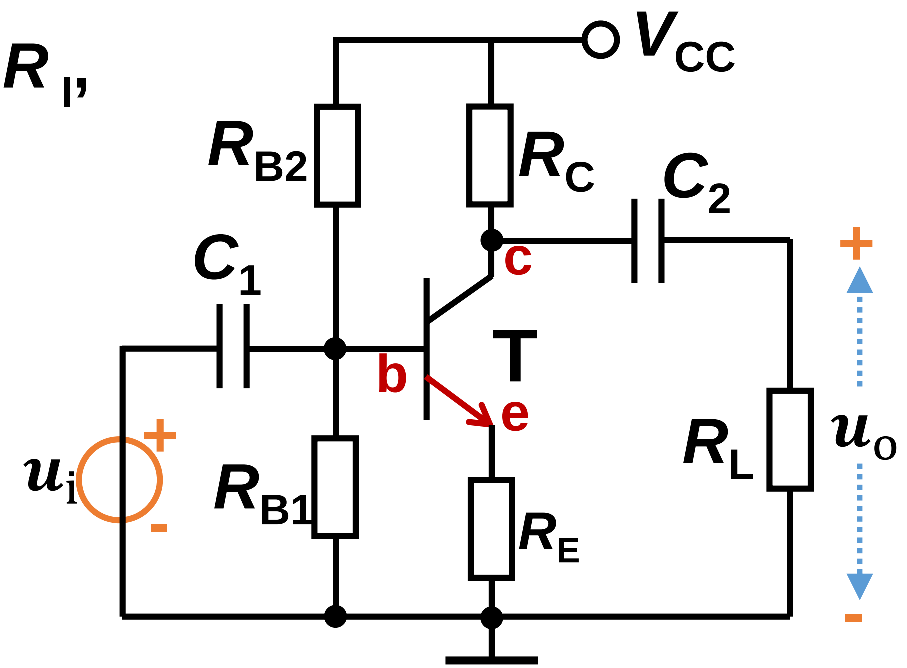


Output resistance: $R_{\rm o} = R_{\rm C} = 2.7\ {\rm k}\Omega$

[Q2] Same parameters as Q1: R_{B2}=39kΩ, R_{B1}=10kΩ, R_C=2.7kΩ, R_E=1kΩ, R_L=5.1kΩ and V_{CC}=15V, transistor has β =100, $r_{bb'}$=300Ω and U_{BEQ}=0.7V, assume capacitors C_1 and C_2 have no resistance to AC signals, ask：

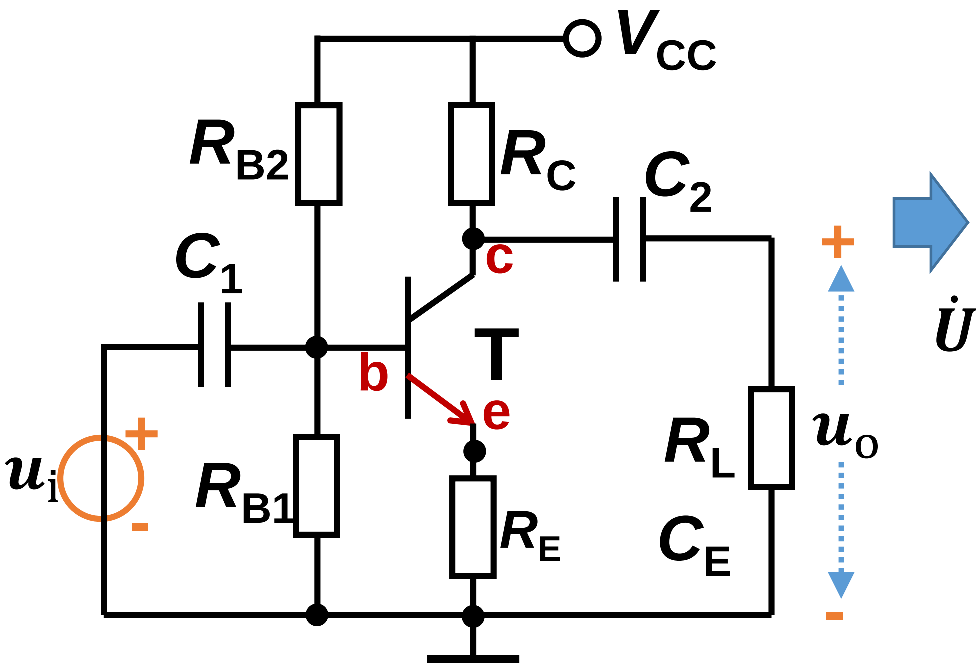
(1) Parameters of quiescent working point Q;

(2) Dynamic parameters A_u, R_I, R_o.

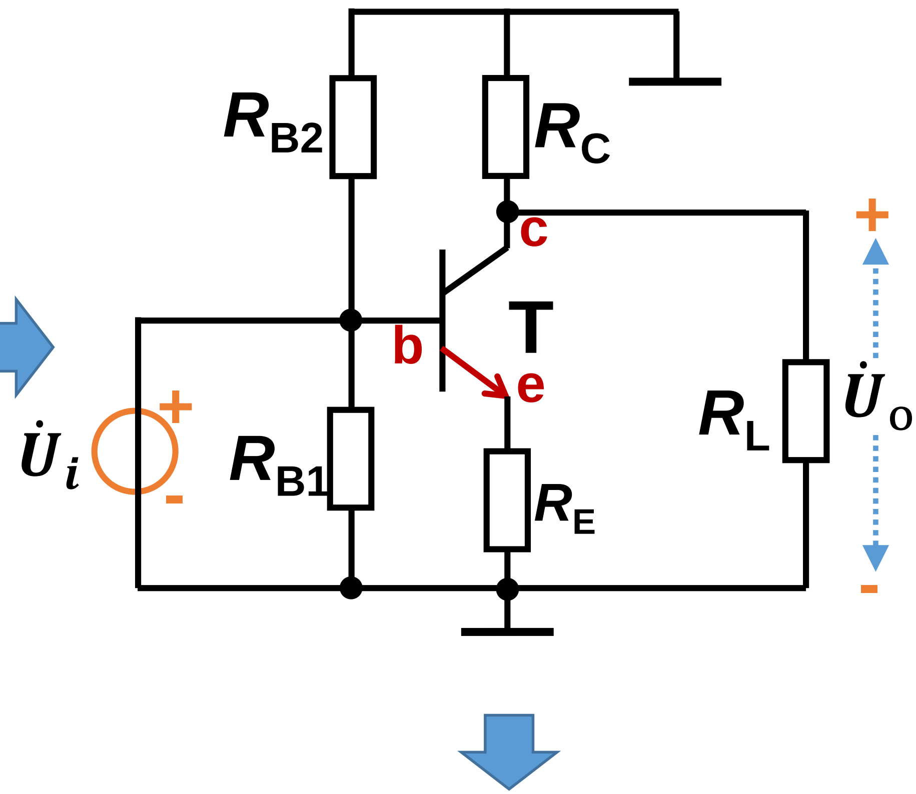


Ans: (1) The DC circuit in Question 2 is the same as Question 1. So the quiescent working point is the same.

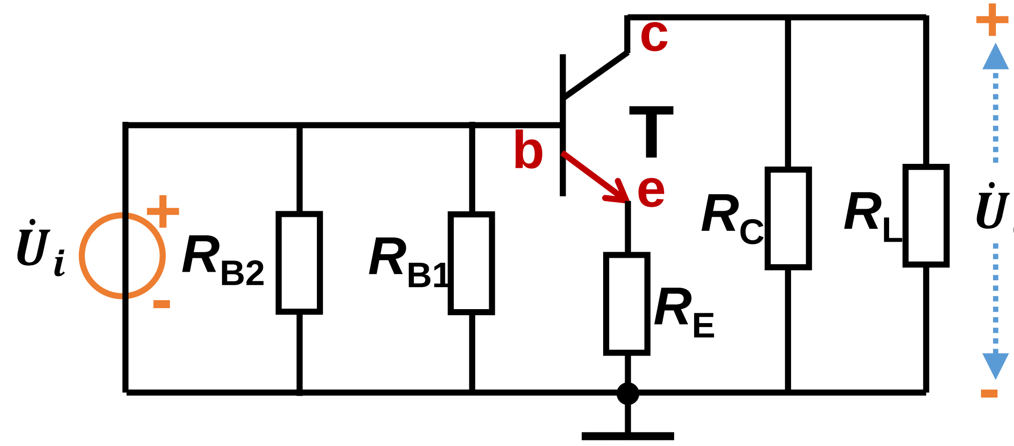
(2) Answer:



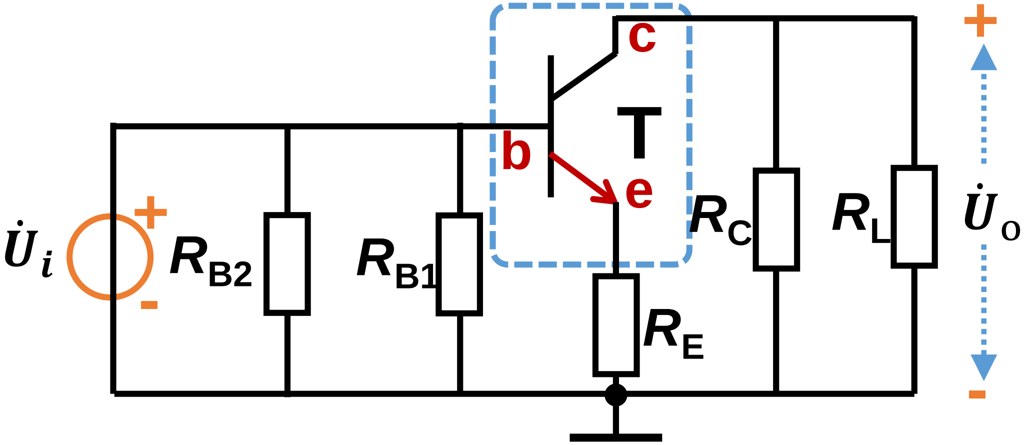
V_{CC}
R_{B2}
R_C
C_2
C_1
c
b
T
e
u_i
R_{B1}
R_E
R_L
C_E
u_o

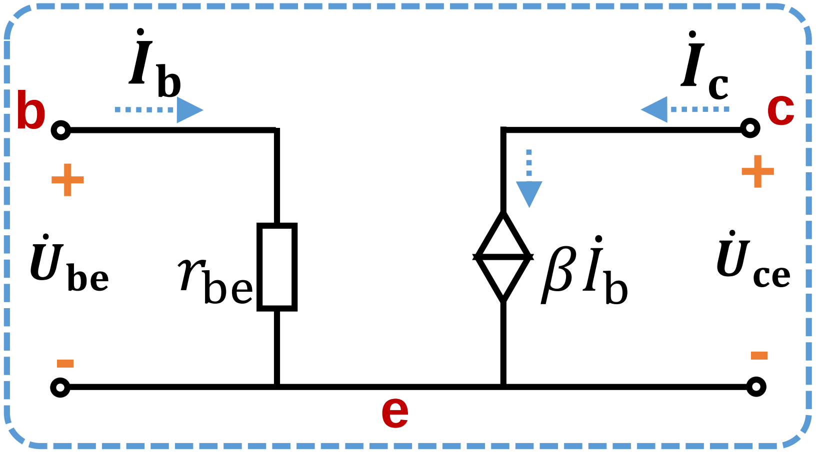


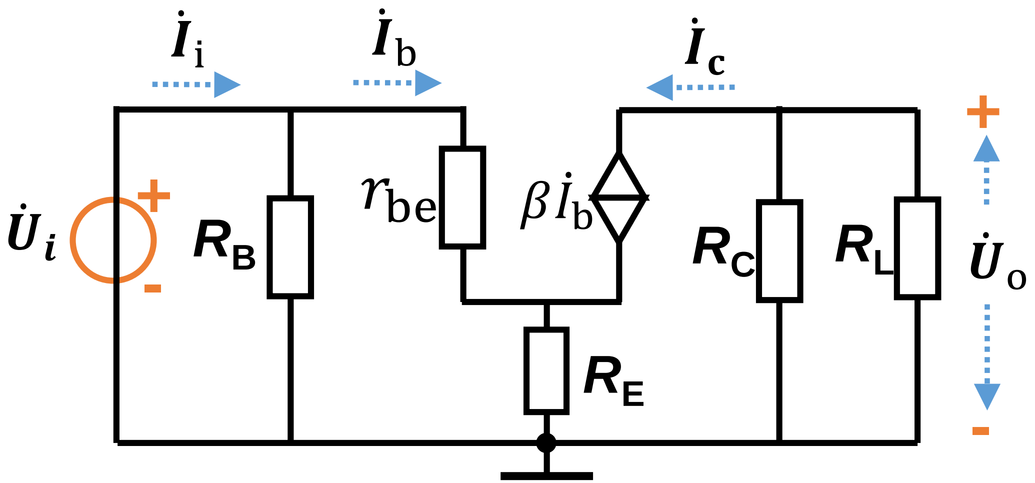
R_{B2}
R_C
c
b
T
e
$\dot{U}_i$
R_{B1}
R_E
R_L
$\dot{U}_o$



c
b
T
e
$\dot{U}_i$
R_{B2}
R_{B1}
R_E
R_C
R_L
$\dot{U}_o$



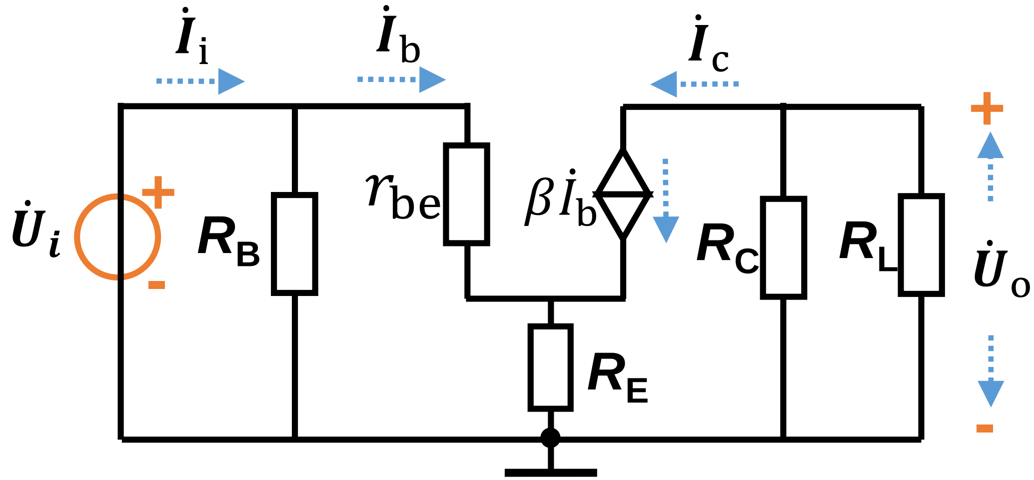




$$R_B = R_{B1}||R_{B2} = 7.96\text{k}\Omega$$

$$\boxed{r_{be} = r_{bb'} + \beta\frac{U_T}{|I_{CQ}|}}$$

$$= 300 + 100\frac{0.026}{0.00234}\text{k}\Omega \approx 1.4\ \text{k}\Omega$$



Voltage gain: $\dot{A}_{\mathrm{u}} = \dfrac{\dot{U}_{\mathrm{o}}}{\dot{U}_{\mathrm{i}}}$

$$\dot{U}_{\mathrm{i}} = \dot{I}_{\mathrm{b}} r_{\mathrm{be}} + (\dot{I}_{\mathrm{b}} + \beta \dot{I}_{\mathrm{b}}) R_{\mathrm{E}}$$

$$\dot{U}_{\mathrm{o}} = -\dot{I}_{\mathrm{c}} (R_{\mathrm{C}} || R_{\mathrm{L}}) = -\beta \dot{I}_{\mathrm{b}} (R_{\mathrm{C}} || R_{\mathrm{L}})$$

$$\dot{A}_{\mathrm{u}} = -\frac{\beta (R_{\mathrm{C}} || R_{\mathrm{L}})}{r_{\mathrm{be}} + (1+\beta) R_{\mathrm{E}}} = -1.72$$

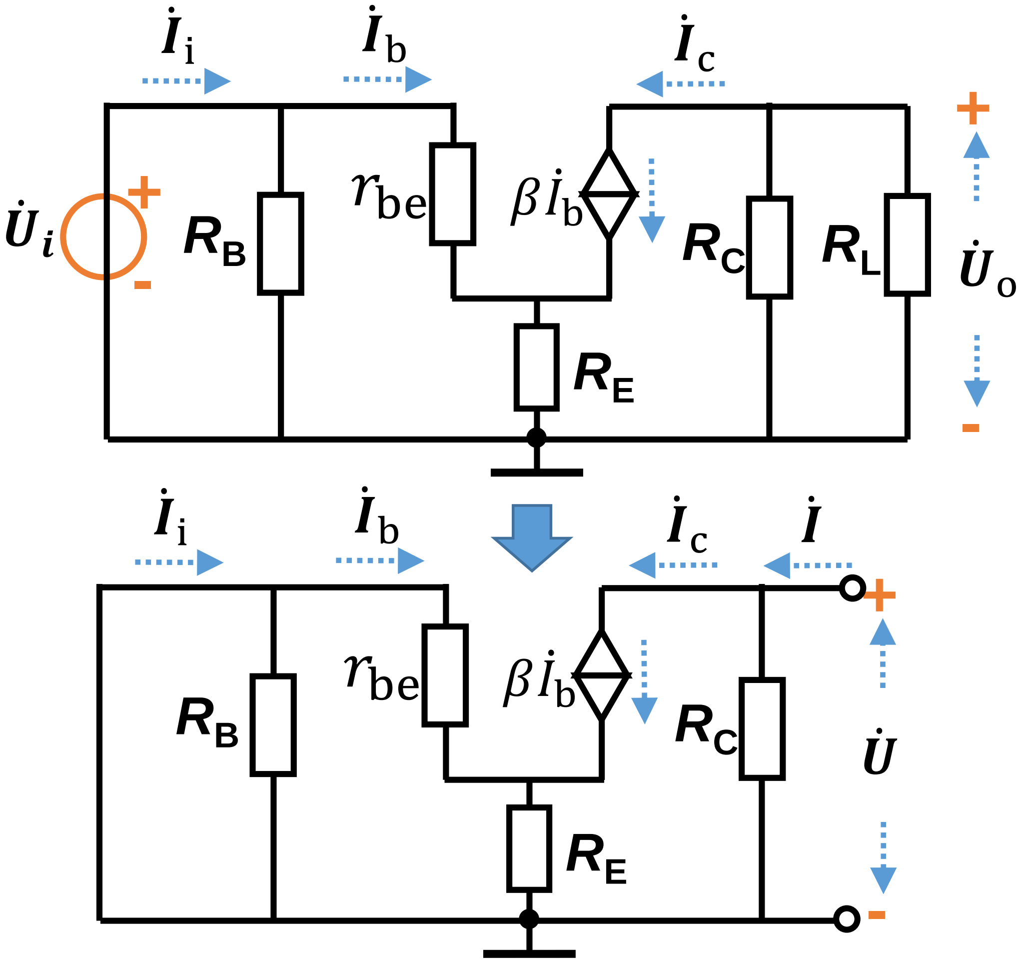
Input resistance: $R_{\mathrm{i}} = \dfrac{\dot{U}_{\mathrm{i}}}{\dot{I}_{\mathrm{i}}}$

$$\dot{U}_{\mathrm{i}} = (\dot{I}_{\mathrm{i}} - \dot{I}_{\mathrm{b}}) R_{\mathrm{B}}$$

$$\dot{U}_{\mathrm{i}} = \dot{I}_{\mathrm{b}} r_{\mathrm{be}} + (\dot{I}_{\mathrm{b}} + \beta \dot{I}_{\mathrm{b}}) R_{\mathrm{E}}$$

$$R_{\mathrm{i}} = R_{\mathrm{B}} \frac{r_{\mathrm{be}} + (1+\beta) R_{\mathrm{E}}}{R_{\mathrm{B}} + r_{\mathrm{be}} + (1+\beta) R_{\mathrm{E}}}$$

$$= 7.38\ \mathrm{k\Omega}$$



Output resistance:

$$R_{\mathrm{o}} = \frac{\dot{U}}{\dot{I}}\bigg|_{R_{\mathrm{L}}=\infty,\dot{U}_{\mathrm{i}}=0}$$

$$\dot{I}_{\mathrm{b}} r_{\mathrm{be}} + \left(\dot{I}_{\mathrm{b}} + \beta\dot{I}_{\mathrm{b}}\right) R_{\mathrm{E}} = 0$$

$$\dot{I}_{\mathrm{b}} = 0$$

$$\dot{I}_{\mathrm{c}} = 0$$

$$R_{\mathrm{o}} = R_{\mathrm{C}}$$

Summary

Common-emitter amplifier

a) Can amplify the voltage signal

b) There is 180^o phase difference between u_o and u_i

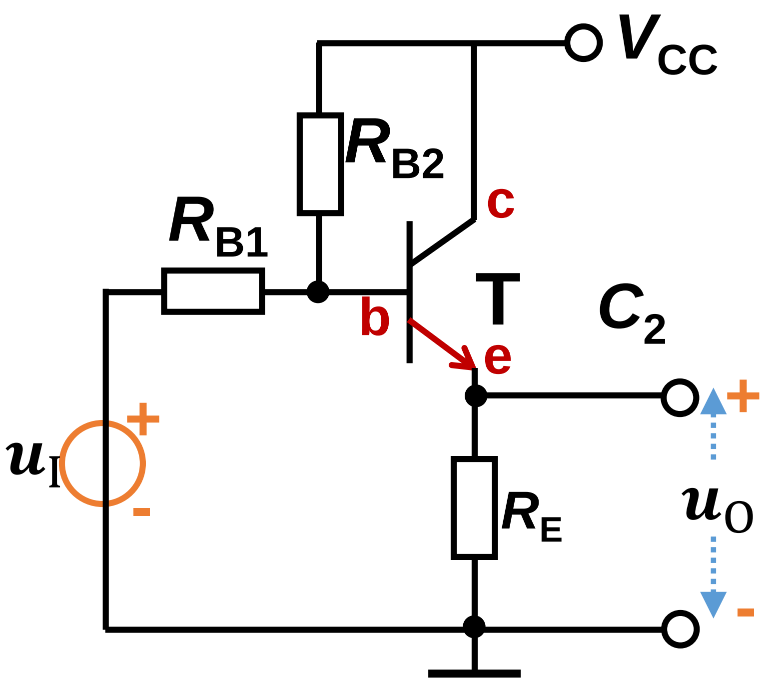
c) Can amplify the current and power
具有电流放大能力和功率放大能力。

d) Low input resistance and high output resistance (compared with common-collector and common-base amplifier).具有低的输入电阻和高的输出电阻(相比于其它接法：共集和共基)。

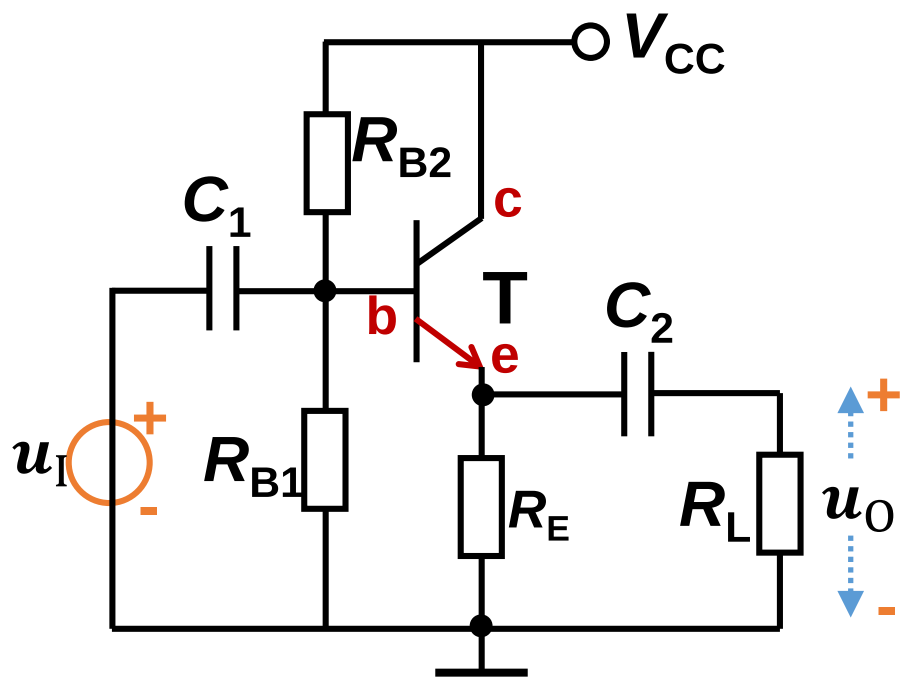
2.4 Common-collector amplifier 共集极放大电路

The output voltage is from emitter（发射极）, is also called emitter follower（射极输出器）

Direct coupling

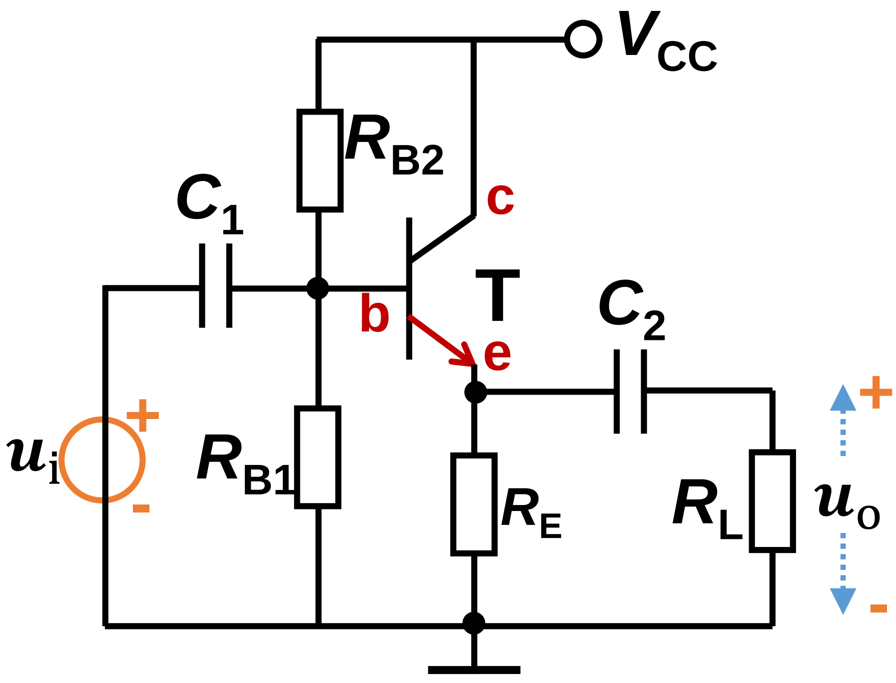


RC coupling

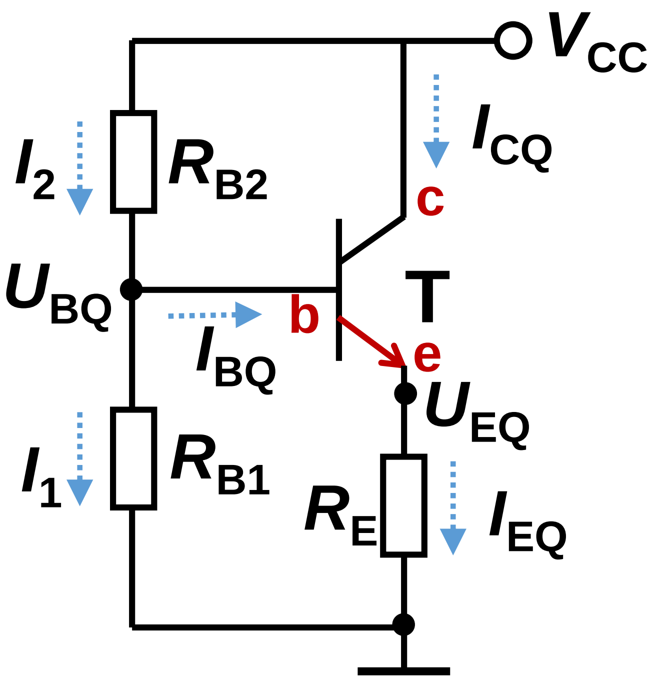


Q: Why is called common-collector?

Analyze DC circuits 静态分析

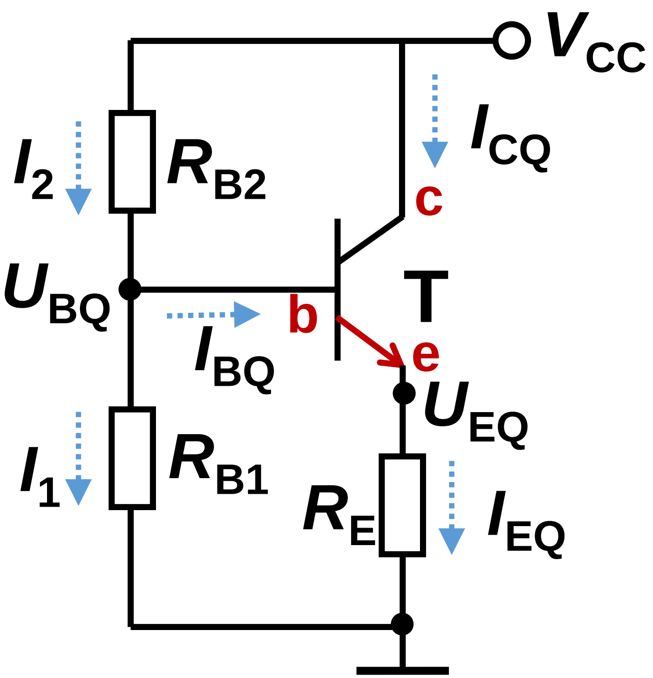


DC circuit



Estimation method估算法

DC circuit



$$I_2, I_1 \gg I_{BQ}$$

$$U_{BQ} \approx \frac{R_{B1}}{R_{B1} + R_{B2}} \cdot V_{CC}$$

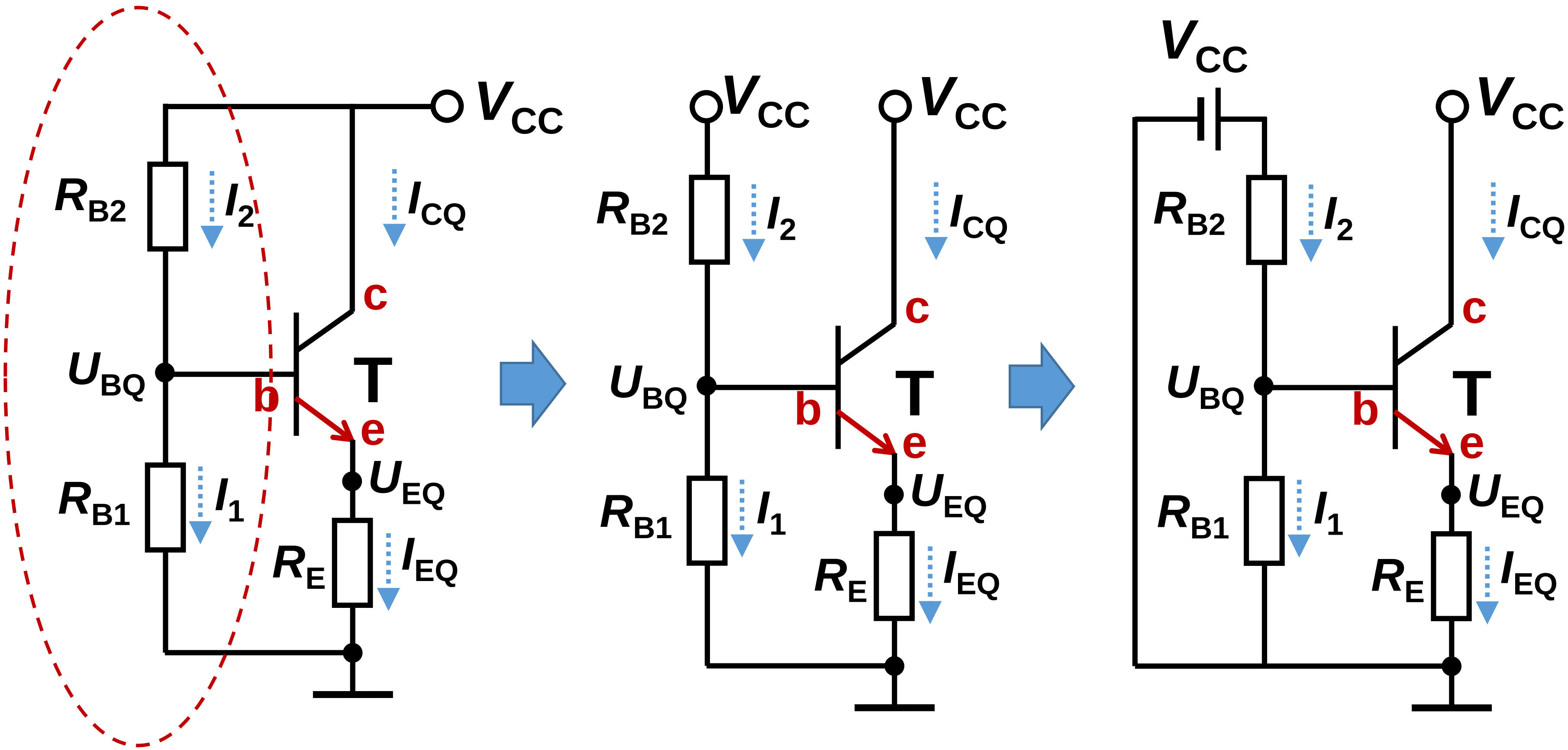
$$I_{EQ} = \frac{U_{EQ}}{R_E} = \frac{U_{BQ} - U_{BEQ}}{R_E}$$

$$I_{BQ} = \frac{I_{EQ}}{\beta + 1}$$

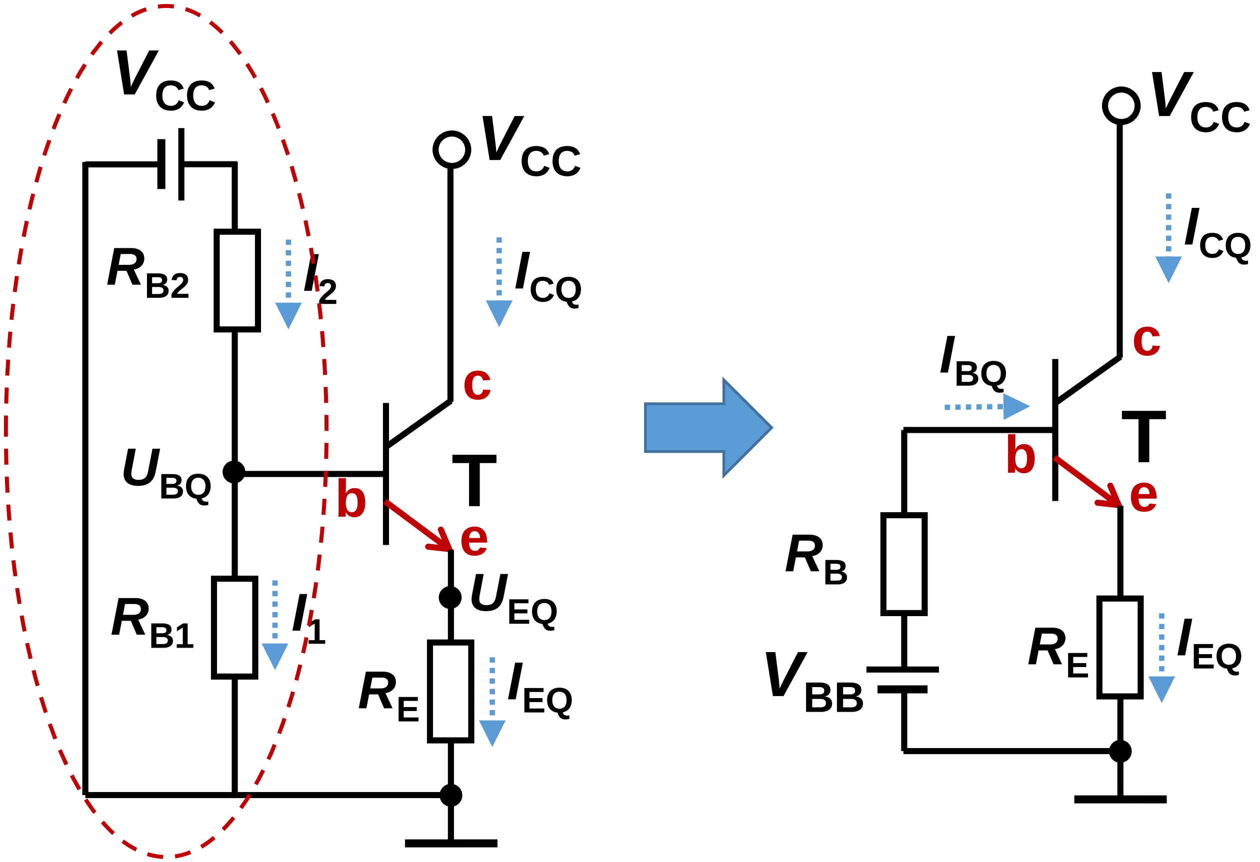
$$I_{CQ} = \beta \frac{I_{EQ}}{\beta + 1}$$

$$U_{CEQ} = V_{CC} - I_{EQ} R_E$$

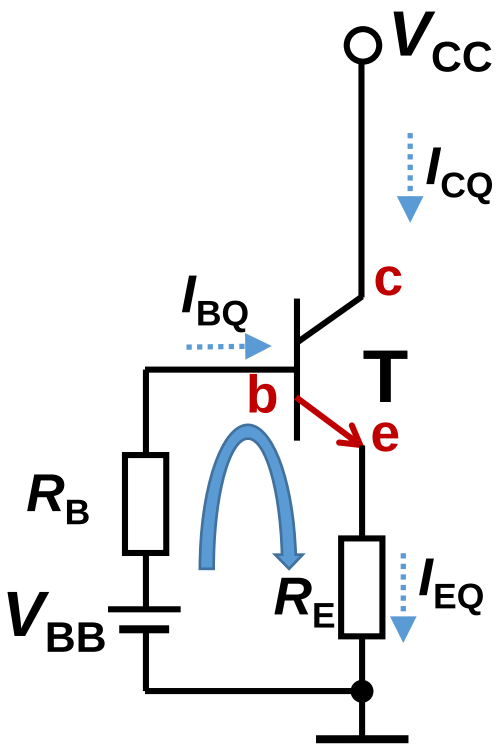
Another approach: Thevenin's theorem 戴维南定理



Another approach: Thevenin's theorem 戴维南定理



$$V_{\mathrm{BB}} = \frac{R_{\mathrm{B1}}}{R_{\mathrm{B1}} + R_{\mathrm{B2}}} \cdot V_{\mathrm{CC}} \qquad R_{\mathrm{B}} = R_{\mathrm{B1}} || R_{\mathrm{B2}}$$



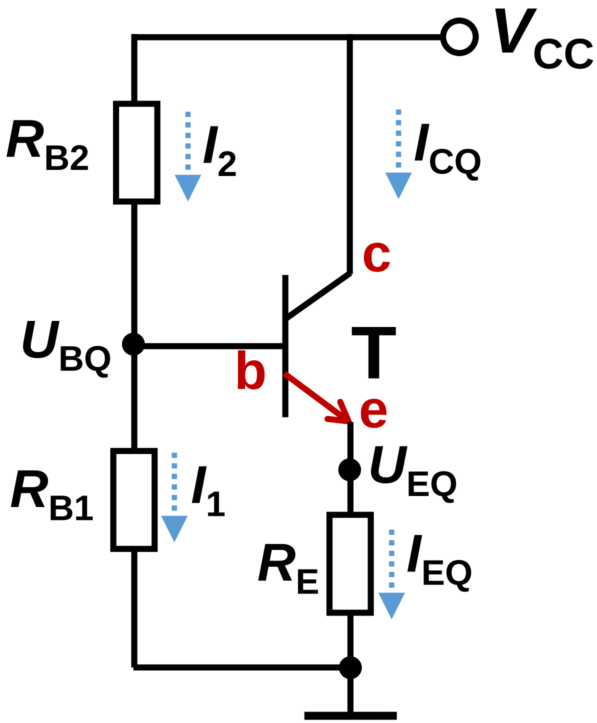
$$V_{BB}=\frac{R_{B1}}{R_{B1}+R_{B2}}\cdot V_{CC}$$

$$R_B=R_{B1}||R_{B2}$$

$$\begin{cases} V_{BB}-I_{BQ}R_B-U_{BEQ}-I_{EQ}R_E=0 \\ I_{EQ}=(1+\beta)I_{BQ} \end{cases}$$

$$I_{BQ}=\frac{V_{BB}-U_{BEQ}}{R_B+(1+\beta)R_E}$$

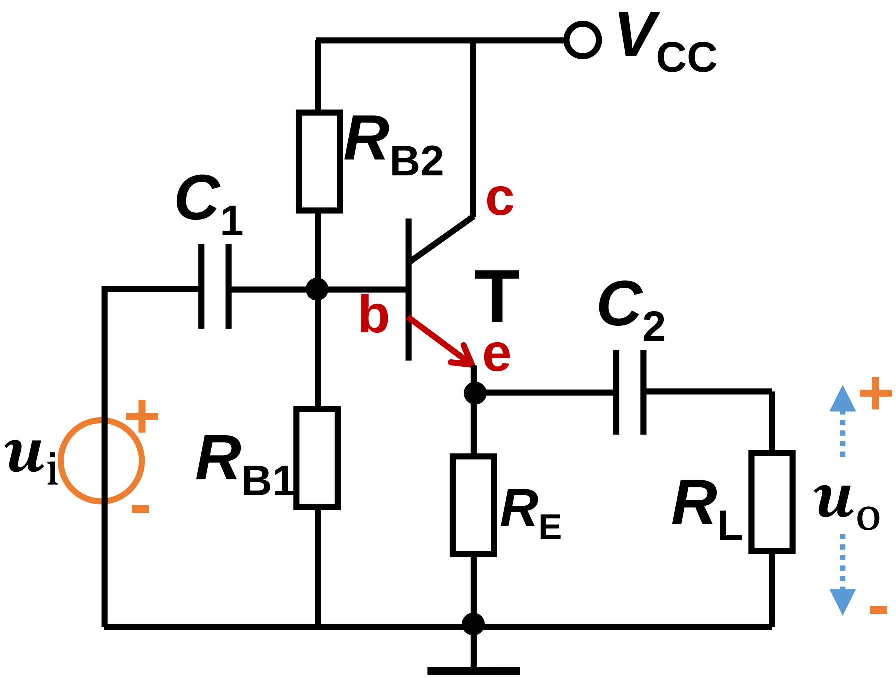
The 3rd approach: 暴力求解法



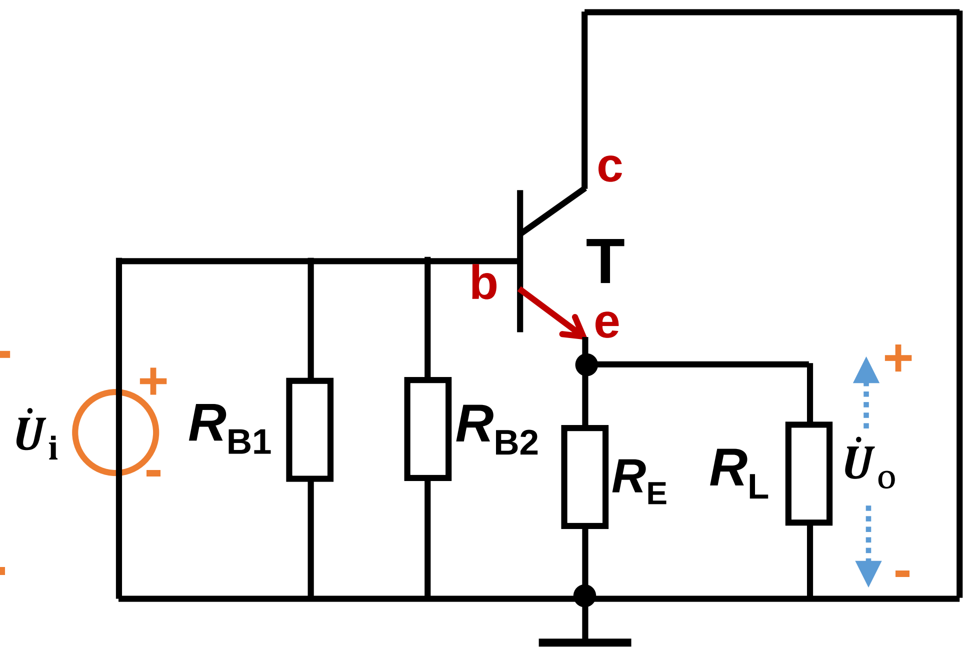
$$\begin{cases} V_{CC} - I_2 R_{B2} - I_1 R_{B1} = 0 \\ U_{BEQ} + I_{EQ} R_E - I_1 R_{B1} = 0 \\ I_2 = I_1 + I_{BQ} \\ I_{EQ} = (1+\beta) I_{BQ} \end{cases}$$

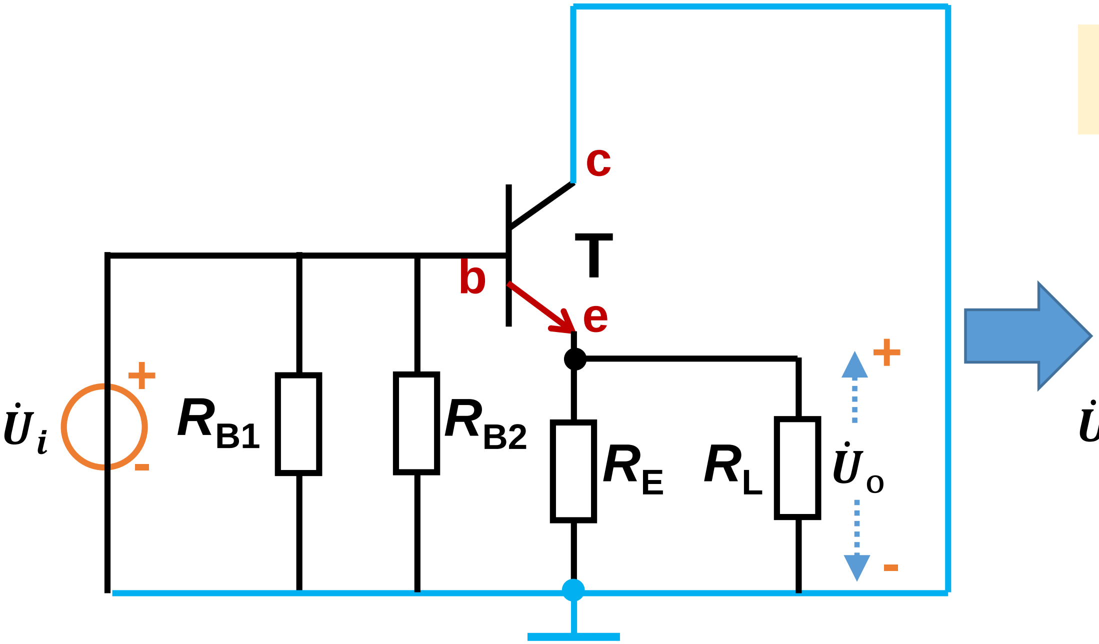
$$I_{BQ} = \frac{V_{BB} - U_{BEQ}}{R_B + (1+\beta) R_E}$$

Dynamic analysis/ Analyze AC circuits 动态分析

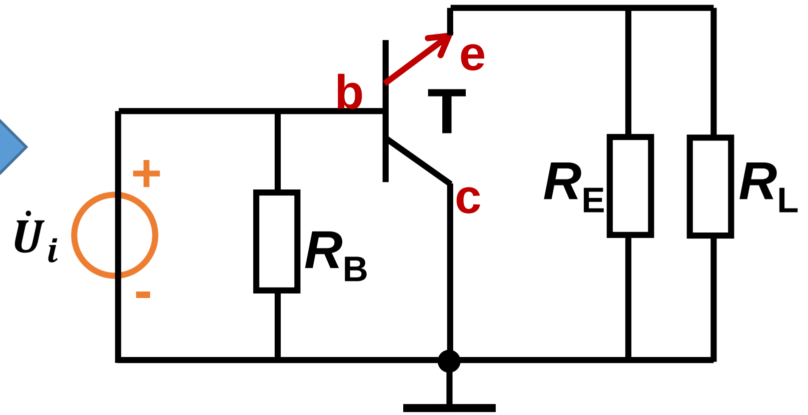


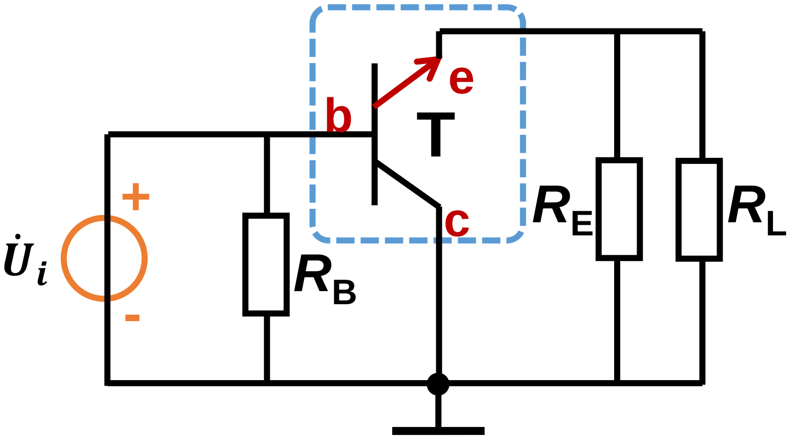
AC circuit

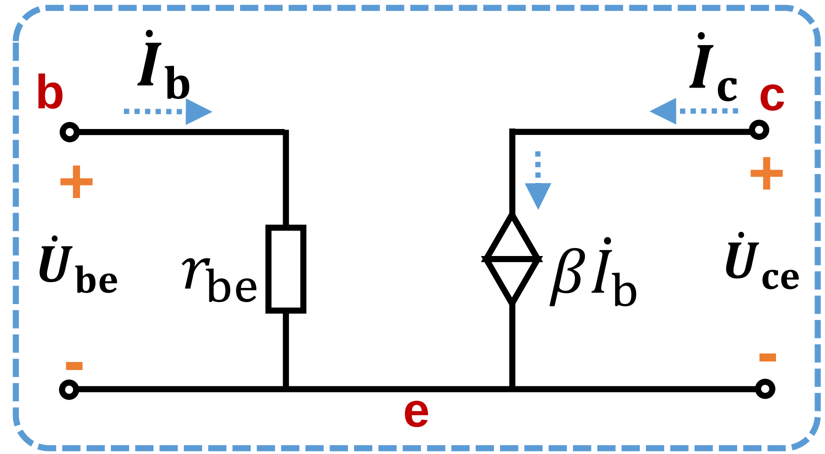




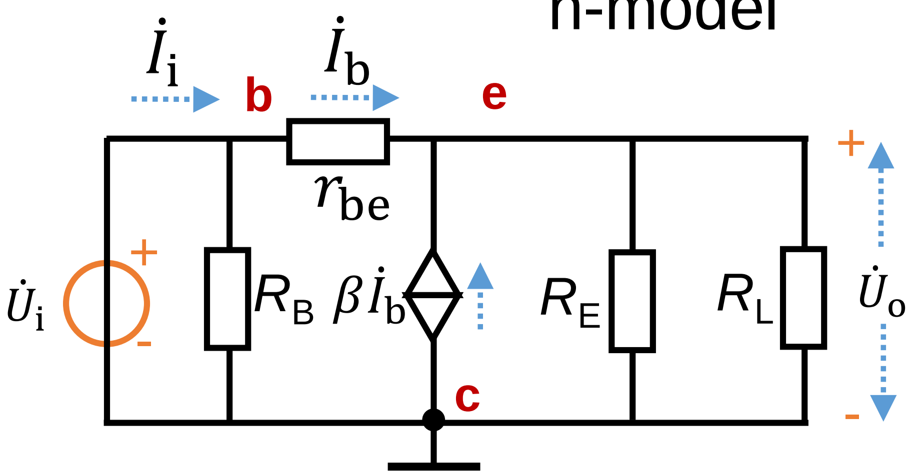
Common-collector!

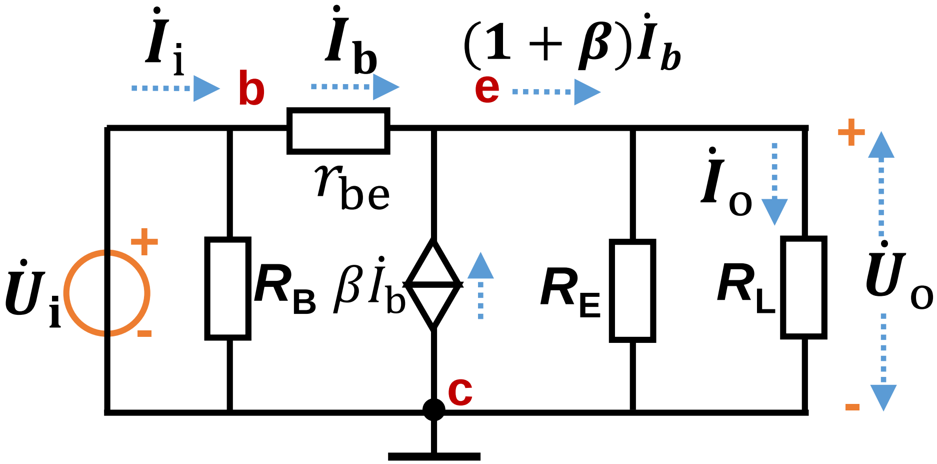



b
e
c
T
$\dot{U}_i$
R_B
R_E
R_L


b
c
e
$\dot{I}_b$
$\dot{I}_c$
$\dot{U}_{be}$
r_{be}
$\beta \dot{I}_b$
$\dot{U}_{ce}$

h-model


$\dot{I}_i$
$\dot{I}_b$
b
e
c
r_{be}
$\dot{U}_i$
R_B
$\beta \dot{I}_b$
R_E
R_L
$\dot{U}_o$



Voltage gain: $\dot{A}_u = \dfrac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_o = (\dot{I}_b + \beta\dot{I}_b)(R_E||R_L)$$

$$\dot{U}_i = \dot{I}_b r_{be} + (\dot{I}_b + \beta\dot{I}_b)(R_E||R_L)$$

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{(1+\beta)(R_E||R_L)}{r_{be} + (1+\beta)(R_E||R_L)}$$

$\dot{A}_u < 1$

Can not amplify voltage signal!

$r_{be} \ll (1+\beta)(R_E||R_L)$ $\quad \dot{A}_u \approx 1$

Current gain: $\dot{A}_i = \dfrac{\dot{I}_o}{\dot{I}_i}$

$$\dot{I}_i = \dot{I}_b + \dot{U}_i/R_B$$

$$= \dot{I}_b + \dot{I}_b \frac{R_i'}{R_B}$$

$$R_i' = r_{be} + (1+\beta)(R_E||R_L)$$

$$\dot{I}_o = \frac{R_E}{R_L + R_E}(\dot{I}_b + \beta\dot{I}_b)$$

$$\dot{A}_i = \frac{R_E}{R_L + R_E}\frac{R_B}{R_i' + R_B}(1+\beta)$$

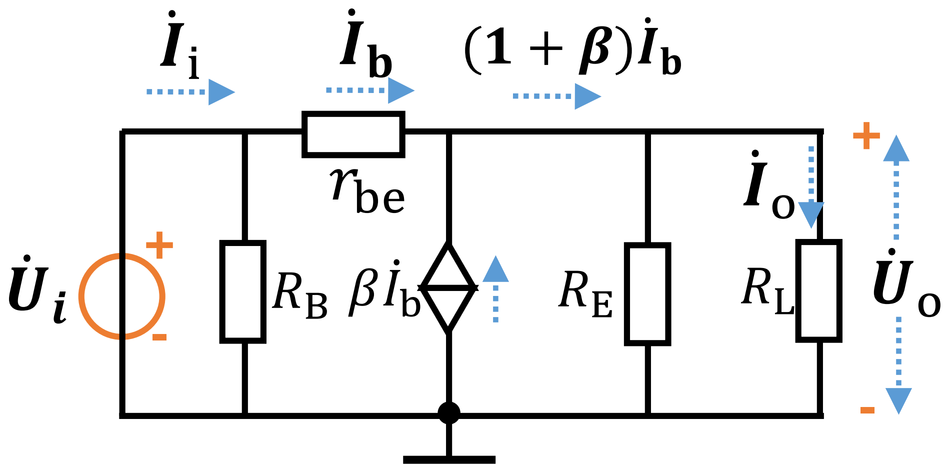
R_E and R_L at same order

R_B and R_i' at same order

$$\dot{A}_i > 1$$

Can amplify current!

Can amplify power!



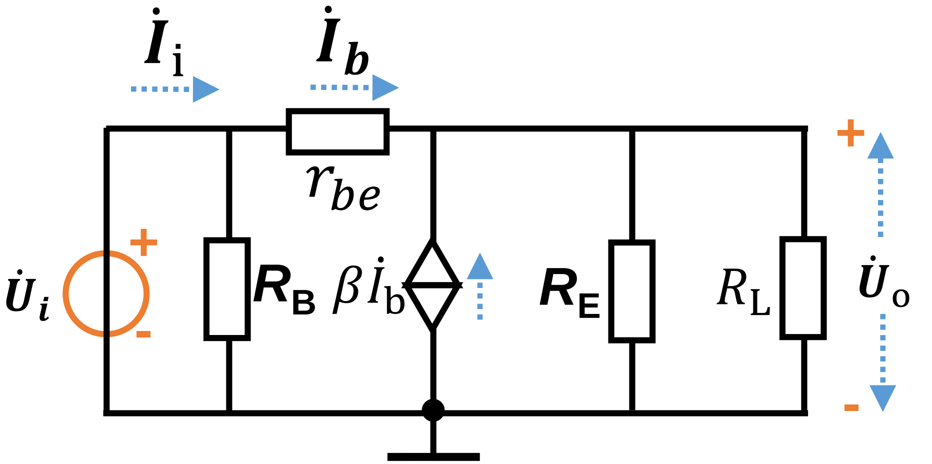
Input resistance: $\boldsymbol{R_i = \dfrac{\dot{U}_i}{\dot{I}_i}}$

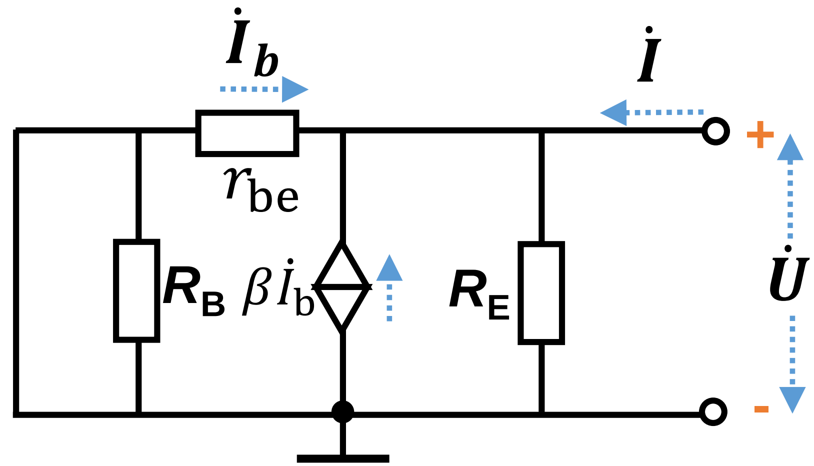
$$\dot{U}_i = \dot{I}_b r_{be} + (\dot{I}_b + \beta\dot{I}_b)(R_E||R_L)$$

$$\dot{I}_i = \dot{I}_b + \dot{U}_i/R_B$$

$$R_i = \frac{\dot{U}_i}{\dot{I}_i} = R_B\frac{r_{be} + (1+\beta)(R_E||R_L)}{R_B + r_{be} + (1+\beta)(R_E||R_L)}$$

$$= R_B||[r_{be} + (1+\beta)(R_E||R_L)]$$





Output resistance: $R_o = \frac{\dot{U}}{\dot{I}}|_{R_L=\infty, \dot{U}_i=0}$

$$\dot{U} = -\dot{I}_b r_{be}$$

$$\dot{U} = (\dot{I} + \dot{I}_b + \beta \dot{I}_b) R_E$$

$$\dot{I} = -\frac{\dot{I}_b r_{be}}{R_E} - \dot{I}_b - \beta \dot{I}_b$$

$$R_o = \frac{\dot{U}}{\dot{I}} = \frac{r_{be}}{\frac{r_{be}}{R_E} + 1 + \beta} \approx \frac{r_{be}}{1+\beta}$$

Output resistance is very small

Summary

Common-collector amplifier

a) $A_u \approx 1$: Can not amplify the voltage signal

b) The phase of u_o and u_i are the same

c) Can amplify the current and power

具有电流放大能力和功率放大能力。

d) High input resistance and low output resistance

Application of Common-collector amplifier

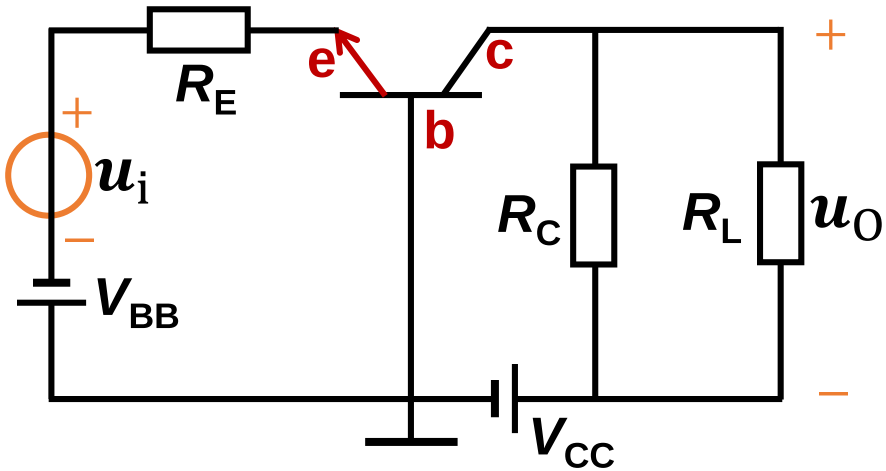
a) Impedance transformation 阻抗变换

Used between high internal resistance source and low resistance load.

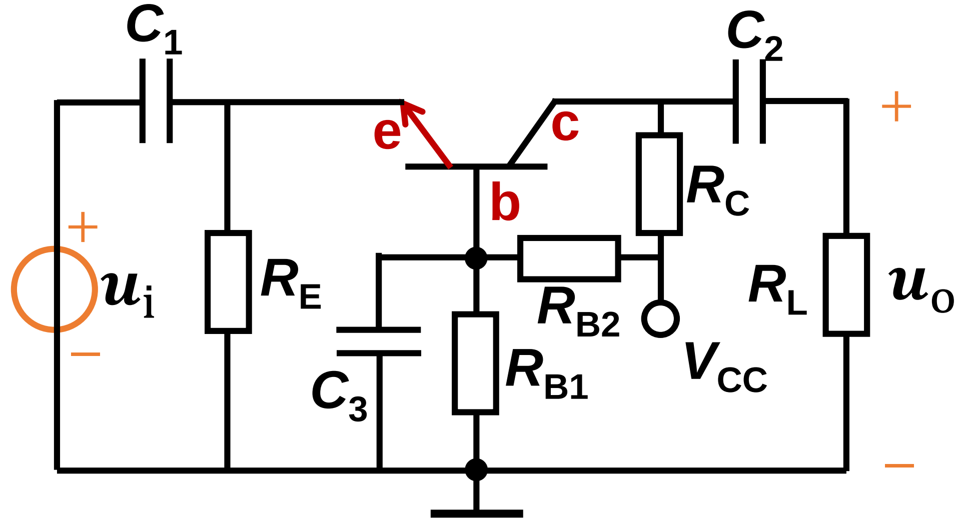
b) Input and output stage of multistage amplifier circuit

2.5 Common-base amplifier　　共基极放大电路

Direct coupling

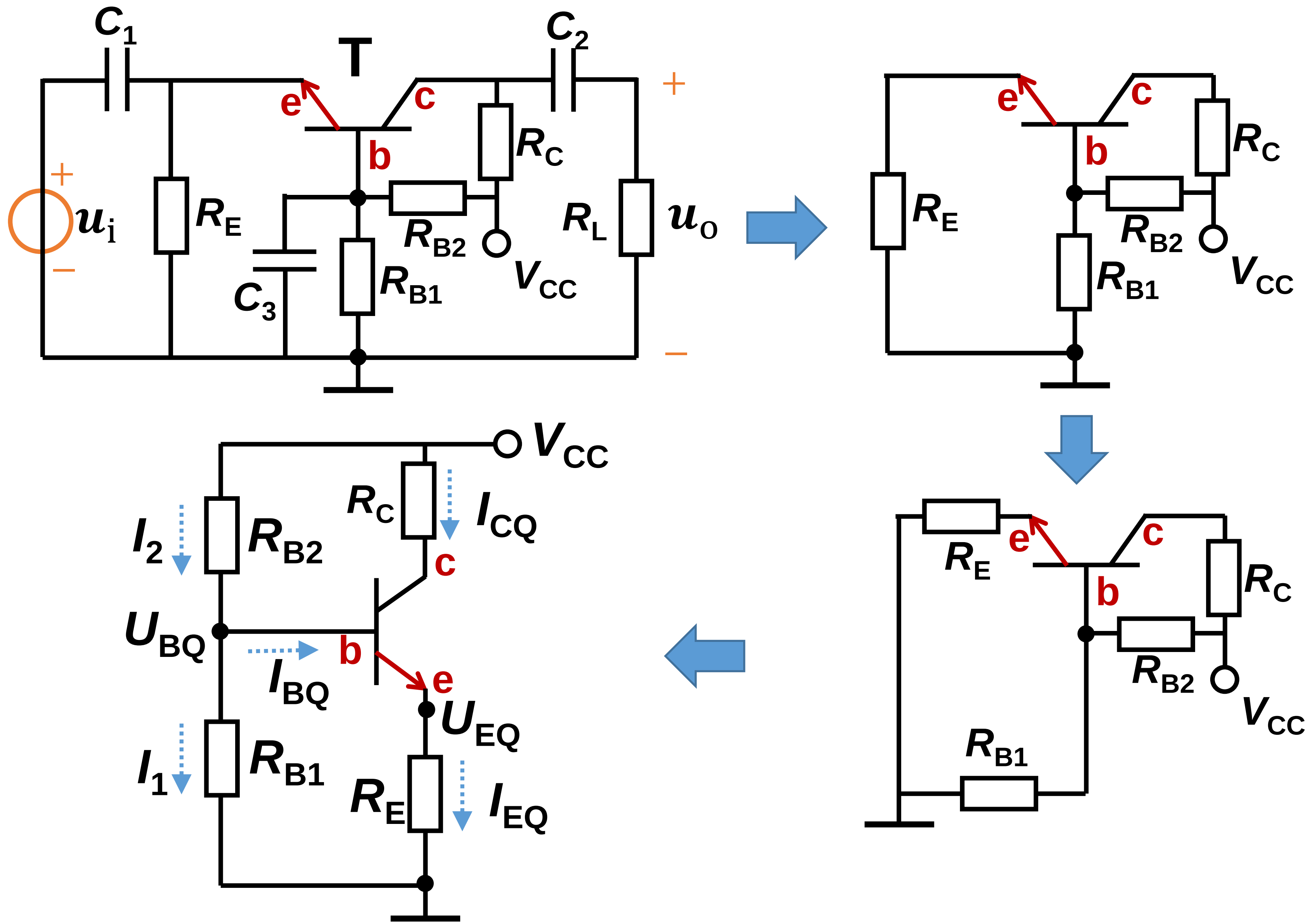


RC coupling

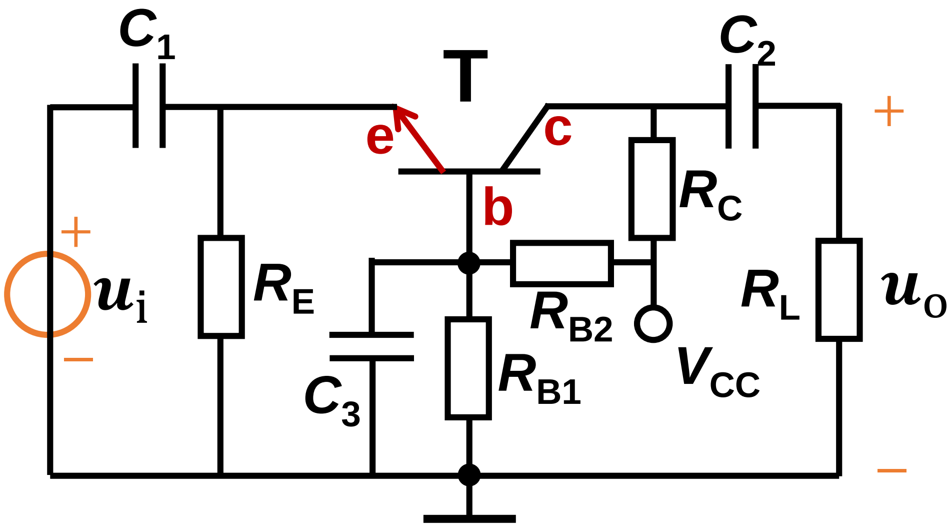


Analyze DC circuits

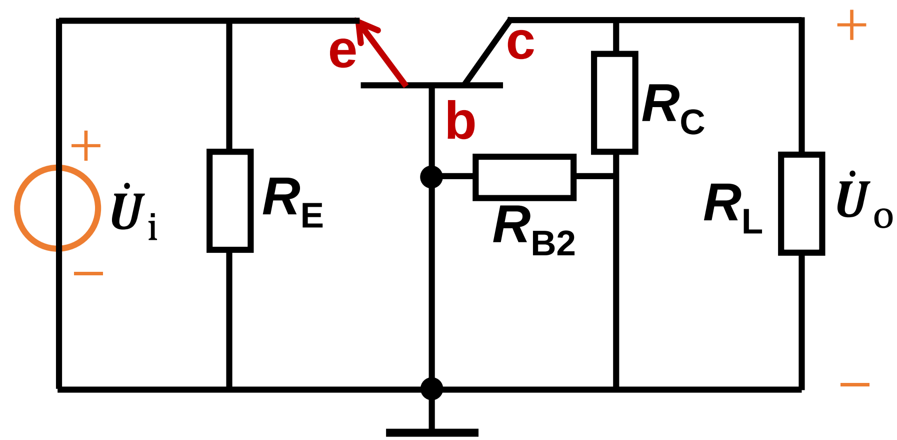
DC circuit

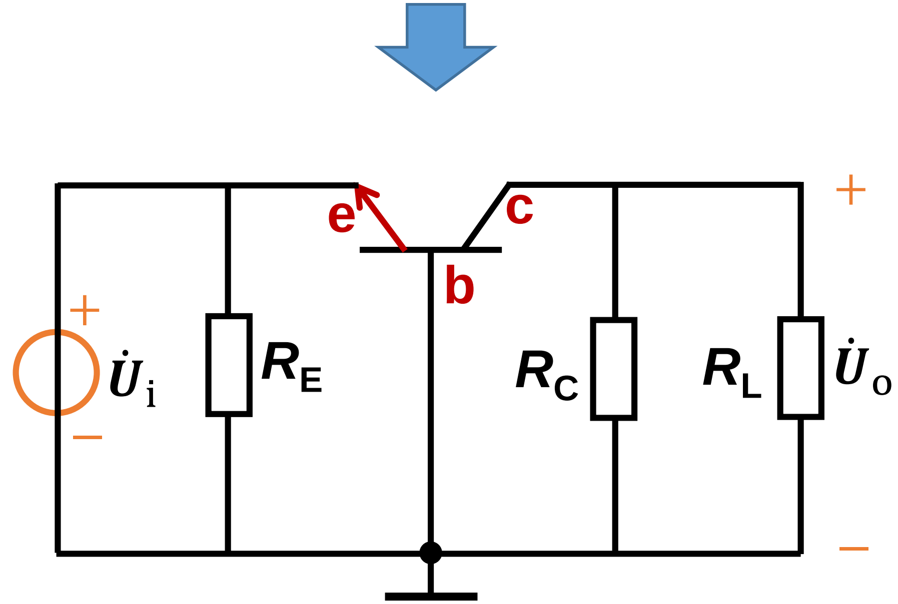


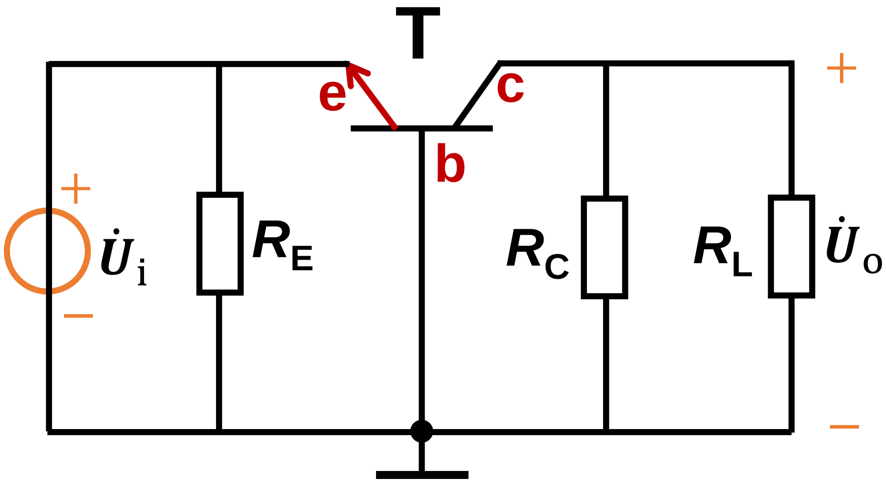
Dynamic analysis

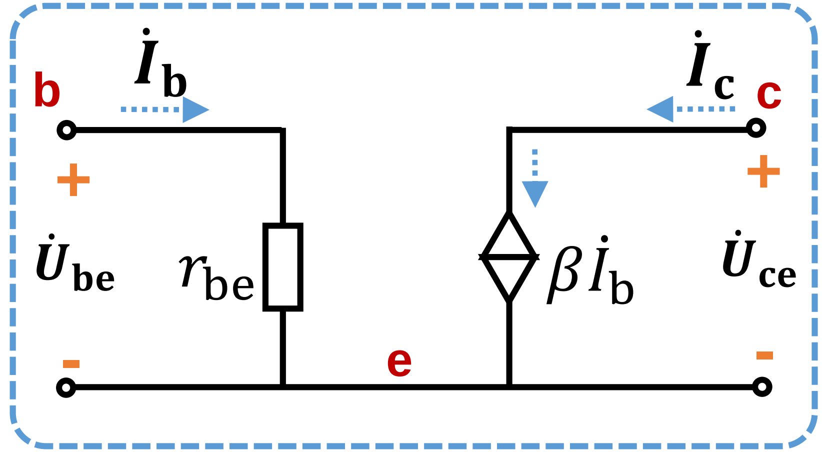


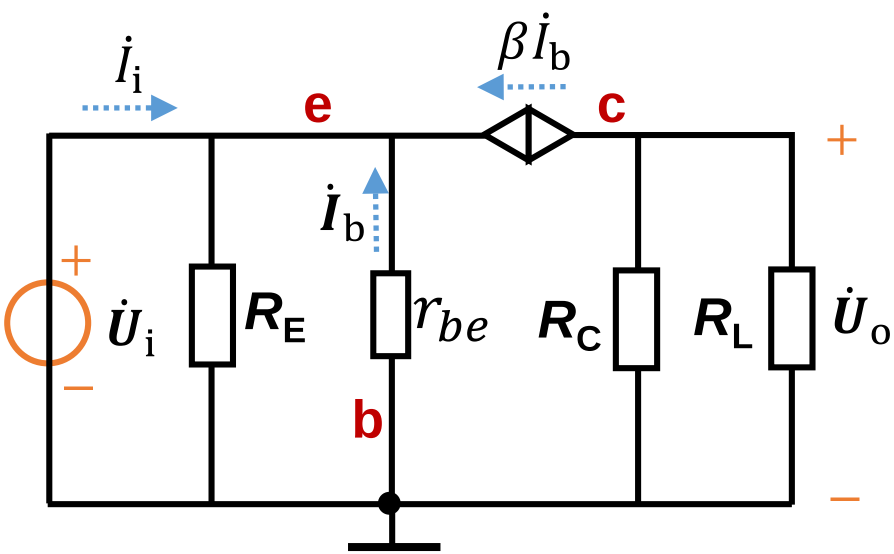
AC circuit











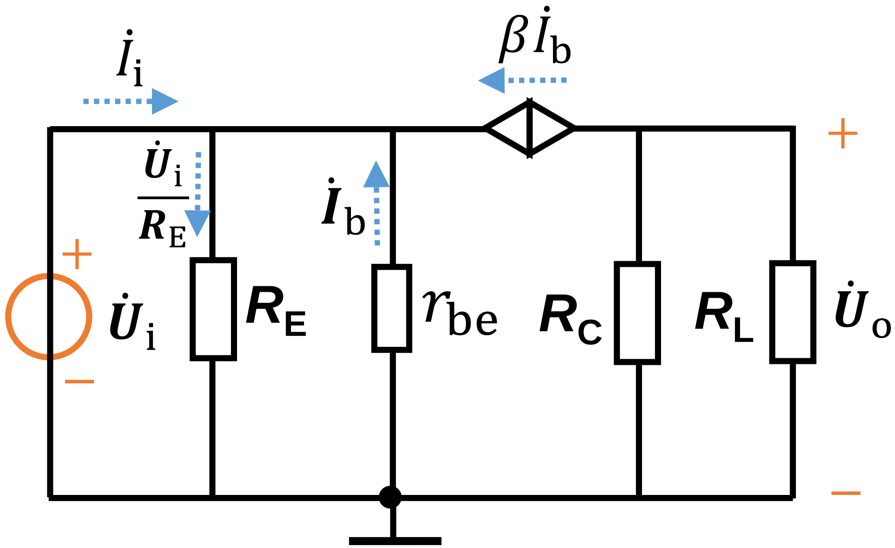
Voltage gain: $\dot{A}_u = \dfrac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_o = -\beta \dot{I}_b (R_C || R_L)$$

$$\dot{U}_i = -\dot{I}_b r_{be}$$

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\beta (R_C || R_L)}{r_{be}}$$

$\dot{A}_u > 1$ **Can amplify voltage**



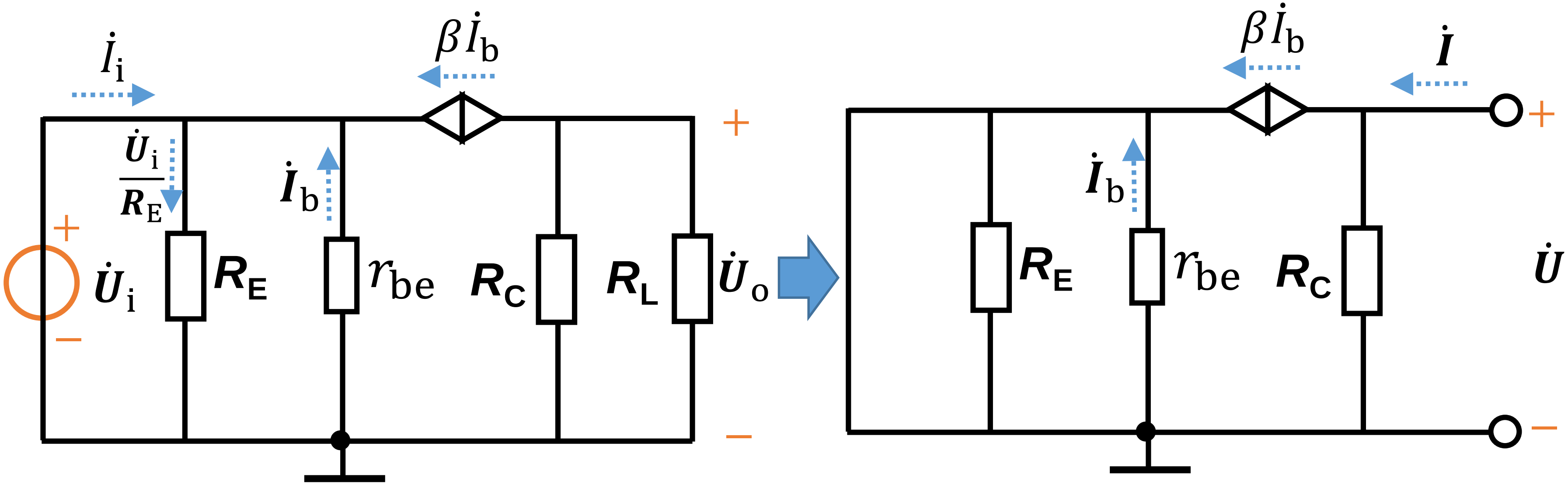
Input resistance: $R_{\mathrm{i}} = \frac{\dot{U}_{\mathrm{i}}}{\dot{I}_{\mathrm{i}}}$

$$\dot{U}_{\mathrm{i}} = -\dot{I}_{\mathrm{b}} r_{\mathrm{be}}$$

$$\dot{I}_{\mathrm{i}} = -\dot{I}_{\mathrm{b}} - \beta \dot{I}_{\mathrm{b}} + \frac{\dot{U}_{\mathrm{i}}}{R_{\mathrm{E}}} = -\dot{I}_{\mathrm{b}} - \beta \dot{I}_{\mathrm{b}} - \frac{\dot{I}_{\mathrm{b}} r_{\mathrm{be}}}{R_{\mathrm{E}}}$$

$$R_{\mathrm{i}} = \frac{r_{\mathrm{be}}}{1 + \beta + \frac{r_{\mathrm{be}}}{R_{\mathrm{E}}}}$$

Small input resistance

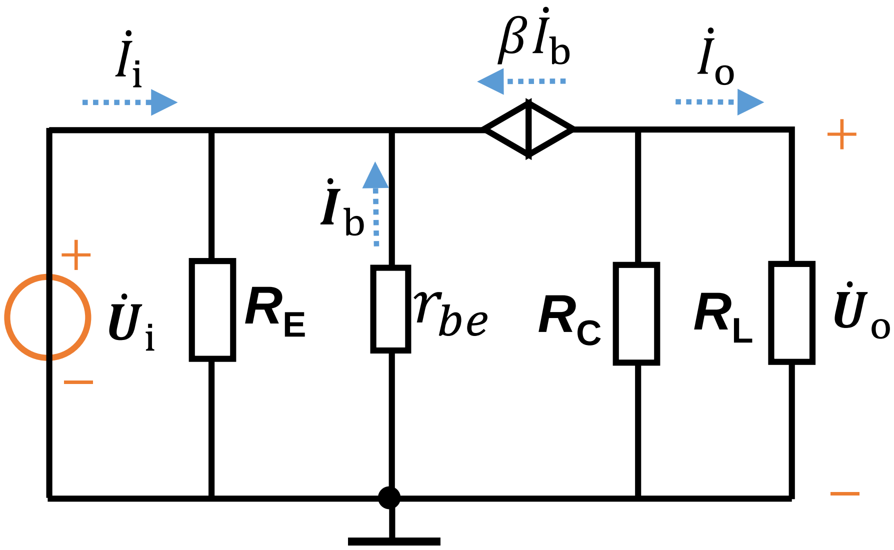


Output resistance:

$$R_o = \frac{\dot{U}}{\dot{I}} \Big|_{R_L=\infty, \dot{U}_i=0}$$

$$R_o = R_c$$

Same as common-emitter



Current gain: $\dot{A}_\mathrm{i} = \dfrac{\dot{I}_\mathrm{o}}{\dot{I}_\mathrm{i}}$

$$\dot{I}_\mathrm{o} = -\frac{R_\mathrm{C}}{R_\mathrm{C} + R_\mathrm{L}}\beta\dot{I}_\mathrm{b}$$

$$\dot{I}_\mathrm{i} = -\dot{I}_\mathrm{b} - \beta\dot{I}_\mathrm{b} - \frac{\dot{I}_\mathrm{b} r_\mathrm{be}}{R_\mathrm{E}}$$

$$\dot{A}_\mathrm{i} = \frac{\dot{I}_\mathrm{o}}{\dot{I}_\mathrm{i}}$$

$$= \frac{1}{1 + R_\mathrm{L}/R_\mathrm{C}} \frac{\beta}{(1+\beta) + r_\mathrm{be}/R_\mathrm{E}}$$

$\dot{A}_\mathrm{i} < 1$

Cannot amplify current!!!

Summary

Common-base amplifier

a) $A_u > 1$: Can amplify voltage signal

b) The phase of u_o and u_i are the same

c) Cannot amplify the current

d) Low input resistance and high output resistance

Compare of common -e, -c and -b amplifier

	Common-e	Common-c	Common-b
$\dot{A}_\mathrm{u}$	High	<1	High
$\dot{A}_\mathrm{i}$ No load resistor	β	$(1+\beta)$	$\alpha=\beta/(1+\beta)$
R_i	Medium	High	Low
R_o	High	Low	High
Bandwidth	Narrow	Medium	Wide

Low frequency amplification circuit

High frequency amplification circuit

Homework 2-2:

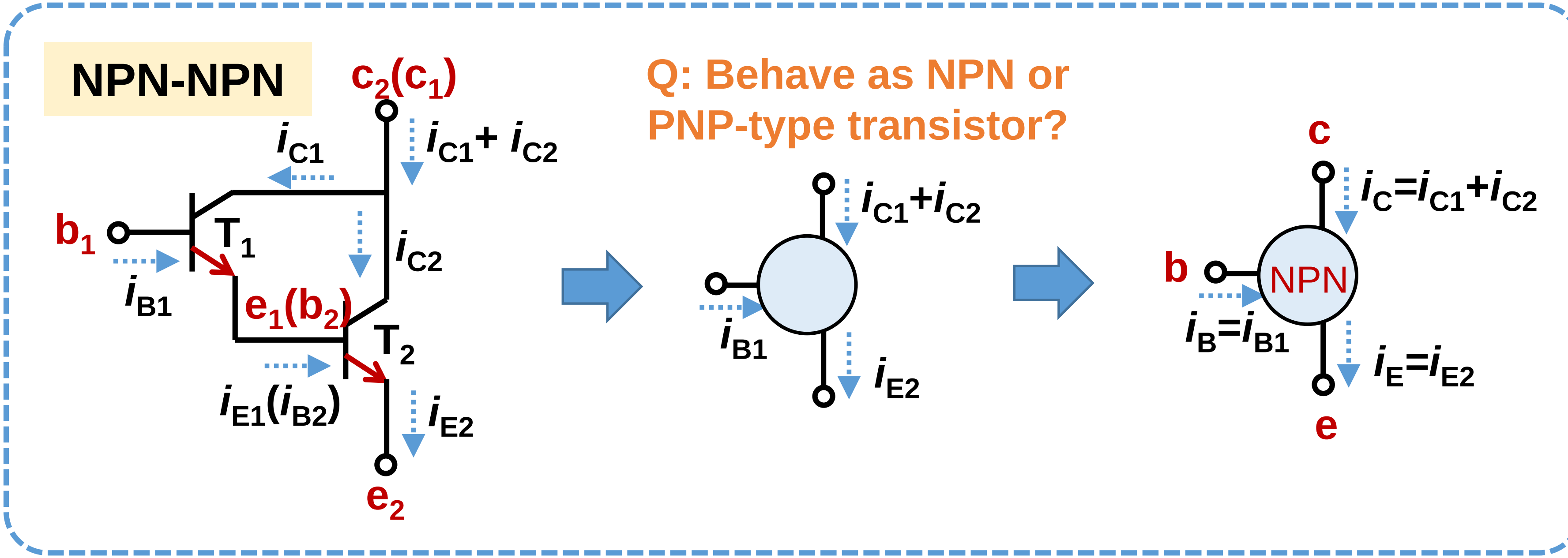
Refer to blackboard and QQ group

Hand in homework 2-1 and 2-2 together!

Deadline: 2022-03-22 11:59pm

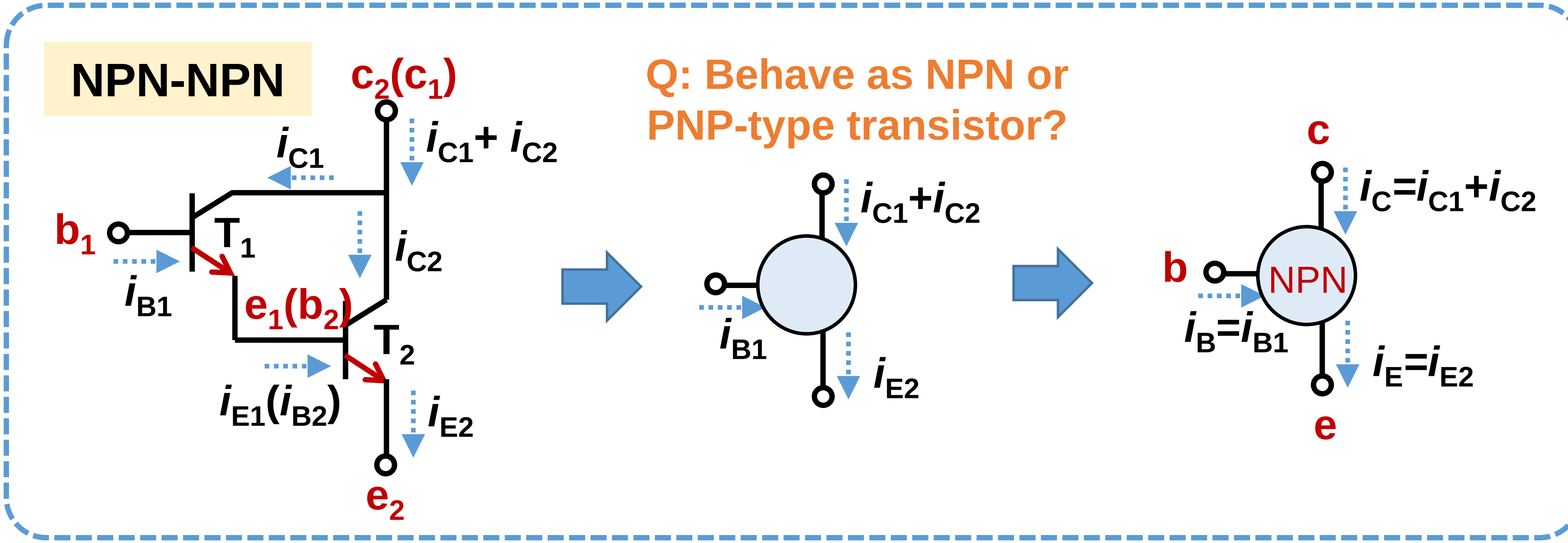
2.6 Composite amplifier 复合管

Connect multiple amplifiers together to improve the voltage gain



NPN-NPN behaves as a **NPN-type** transistor

What's the current amplification factor i_C/i_B?



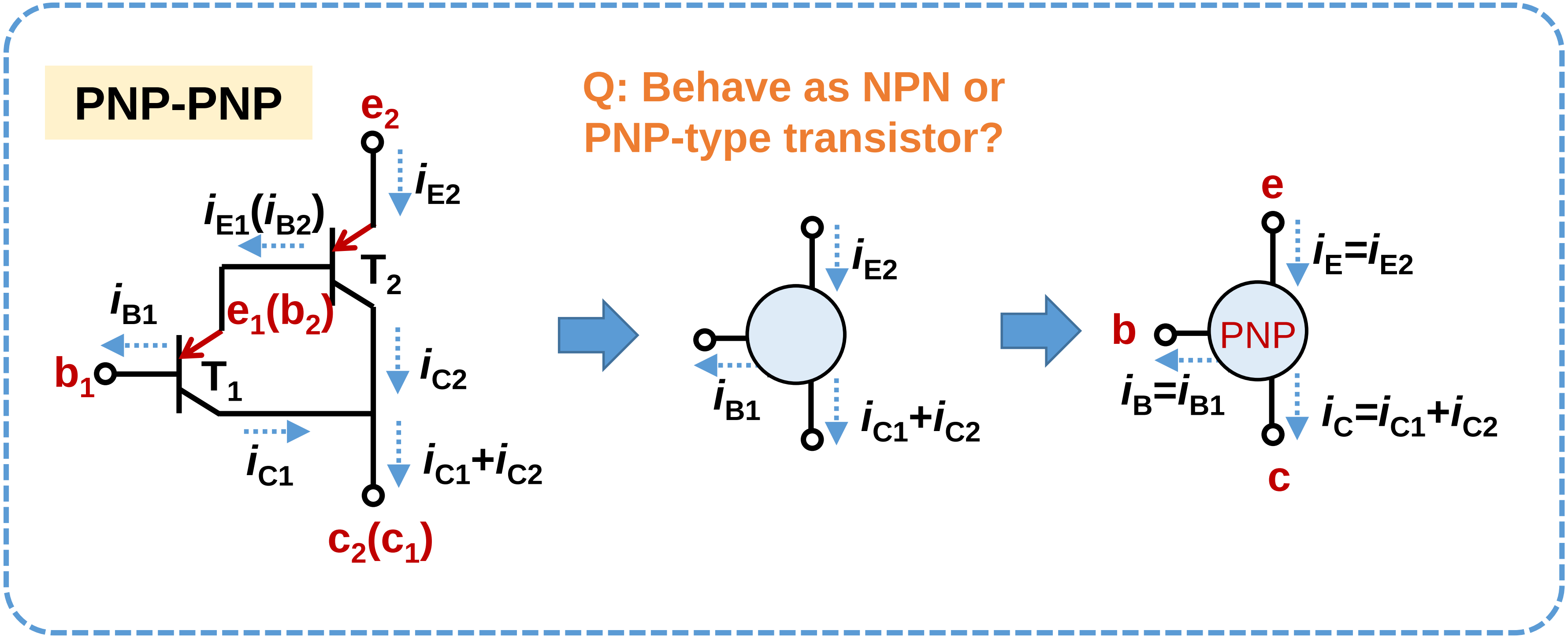
What's the current amplification factor i_C/i_B?

$$i_B = i_{B1} \quad i_C = i_{C1} + i_{C2} = \beta_1 i_{B1} + \beta_2 i_{B2}$$

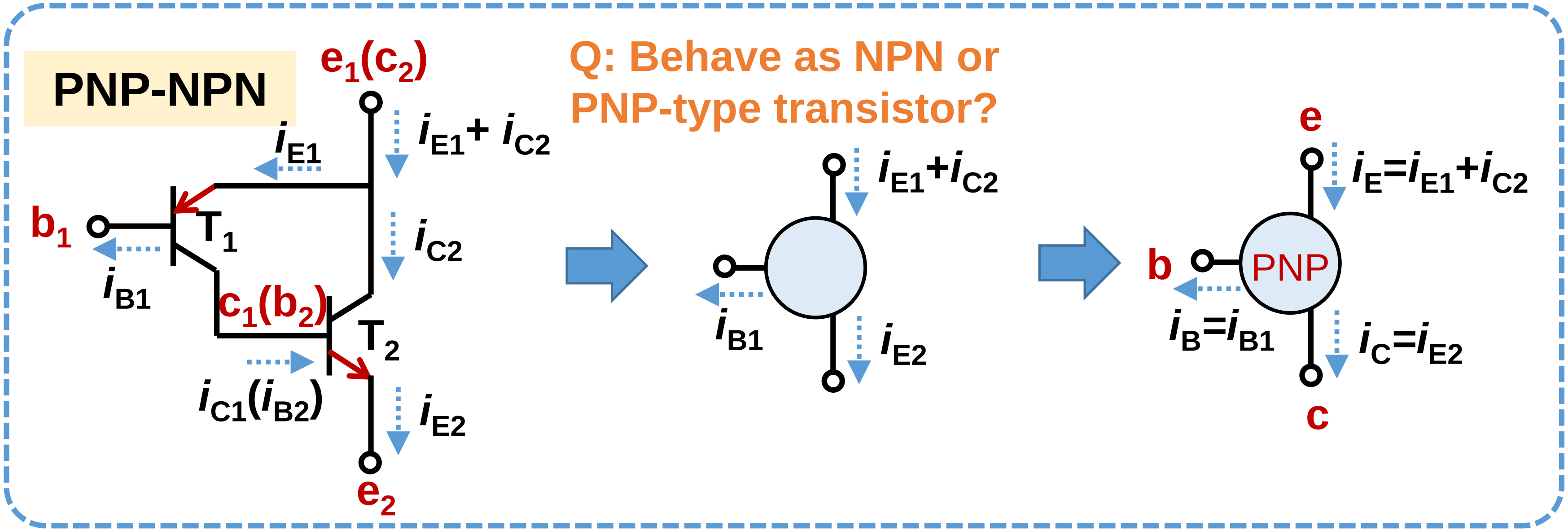
$$i_{B2} = i_{E1} = (\beta_1 + 1) i_{B1}$$

$$i_C = \beta_1 i_{B1} + \beta_2(\beta_1 + 1) i_{B1}$$

$$\boldsymbol{\beta = \beta_1 \beta_2 + \beta_1 + \beta_2}$$



PNP-PNP behaves as a **PNP-type** transistor



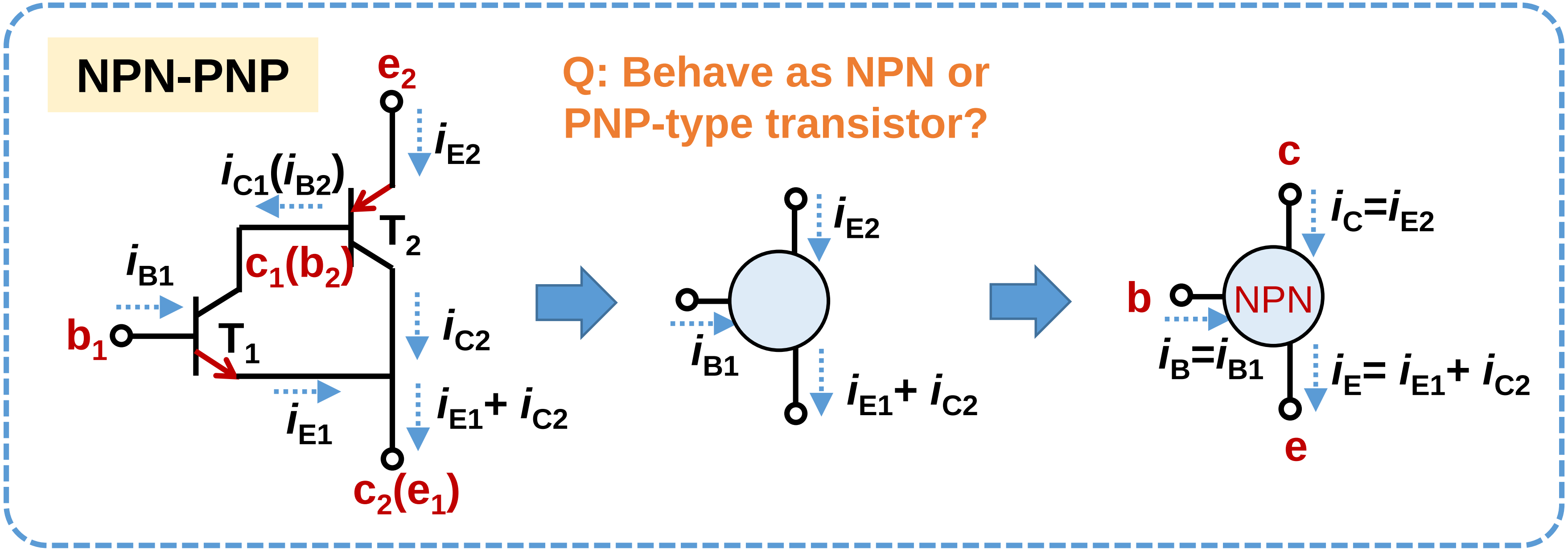
PNP-NPN behaves as a **PNP-type** transistor

The current amplification factor:

$$i_{\mathrm{B}} = i_{\mathrm{B1}} \qquad i_{\mathrm{C}} = i_{\mathrm{E2}} = (\beta_2 + 1)i_{\mathrm{B2}} \qquad i_{\mathrm{B2}} = i_{\mathrm{C1}} = \beta_1 i_{\mathrm{B1}}$$

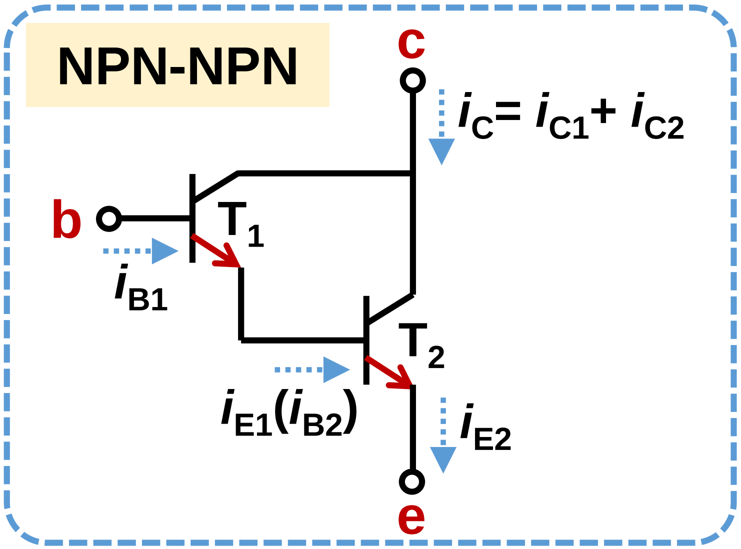
$$i_{\mathrm{C}} = \beta_1(\beta_2 + 1)i_{\mathrm{B1}} = \beta_1(\beta_2 + 1)i_{\mathrm{B}}$$

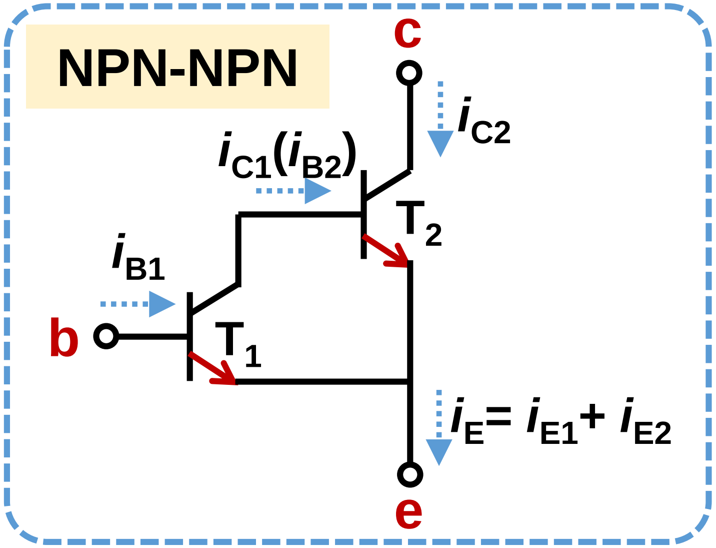
$$\boldsymbol{\beta = \beta_1(\beta_2 + 1)}$$



NPN-PNP behaves as a **NPN-type** transistor

The type of composite transistor is determined by T_1





Q: Can the configuration of right figure amplify the AC signal?

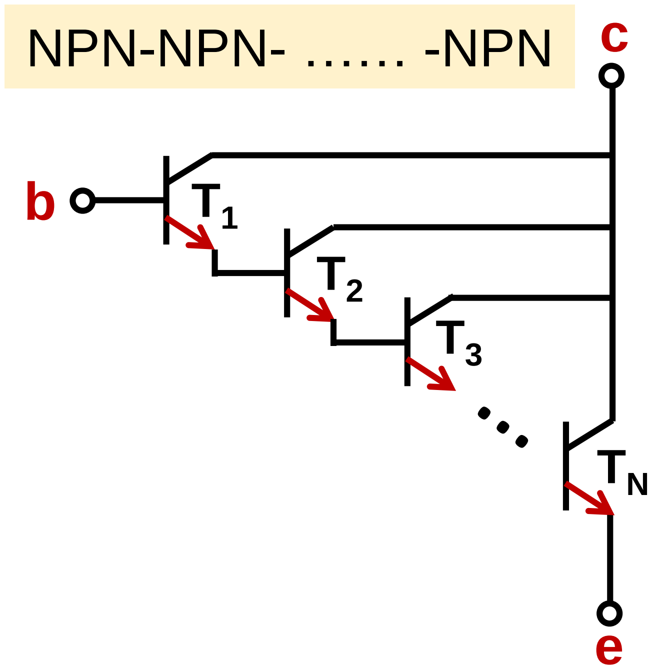
$$i_B = i_{B1} \quad i_C = i_{C2} = \beta_2 i_{B2} \quad i_{B2} = i_{C1} = \beta_1 i_{B1} \quad i_C = \beta_1 \beta_2 i_B$$

The operation region of Transistor T1 is close to saturation region. **Dynamic range is very small**.

Base principle to build a composite transistor

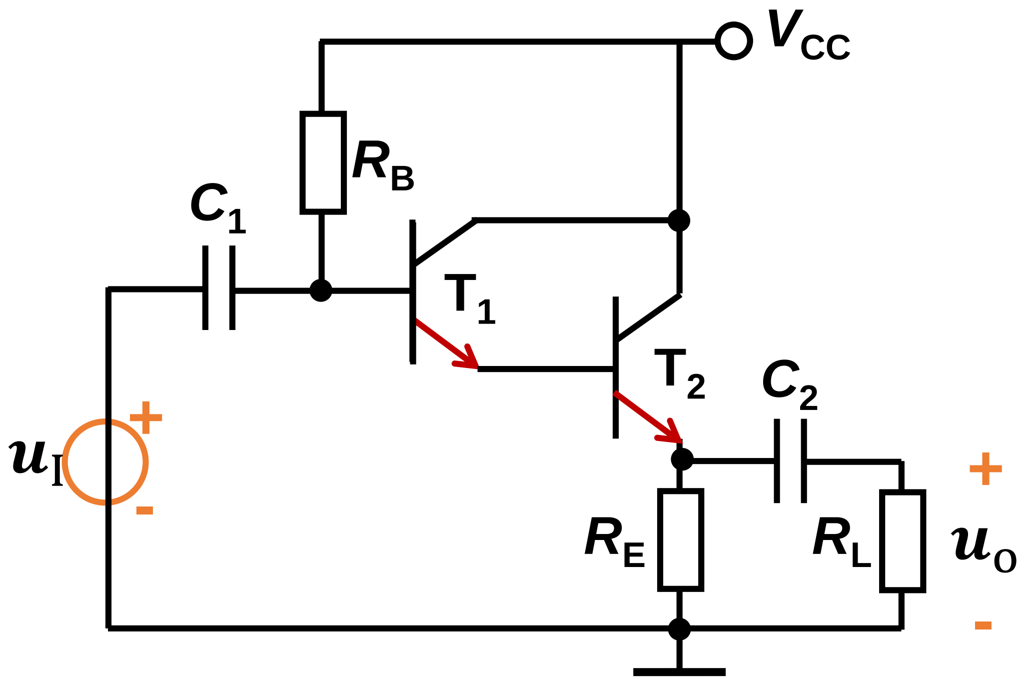
1. Each transistor needs to work in the amplification region

2. Connect the emitter/collector of T_1 to the base of T_2, connect the emitter/collector of T_2 to the base of T_3 …

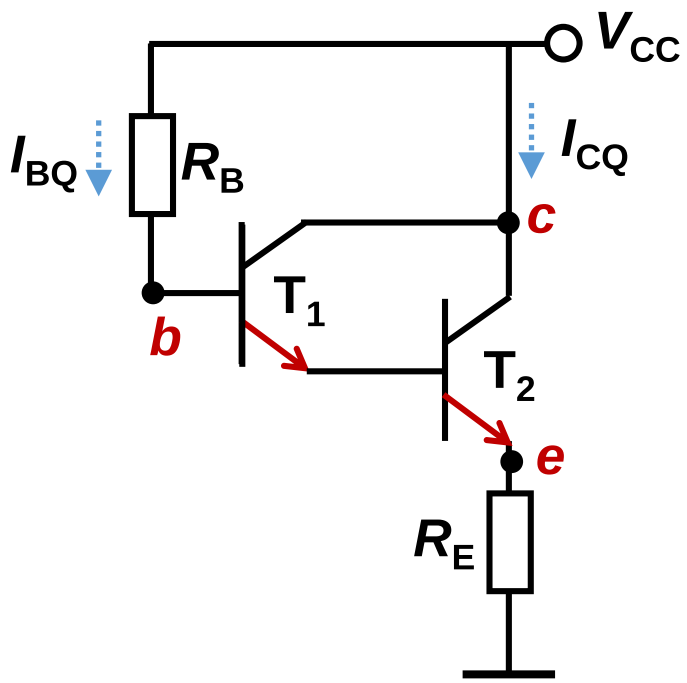


2.6.2 Composite amplifier circuit

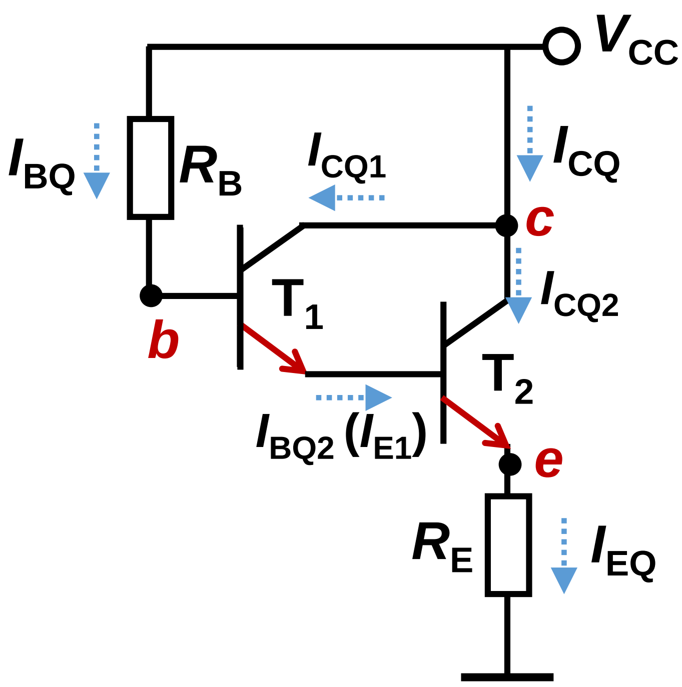
Common-collector composite amplifier circuit



DC circuit



Analyze DC circuits



Input circuit equation

$$V_{CC} = I_{BQ} R_B + U_{BEQ} + (I_{BQ} + I_{CQ}) R_E$$

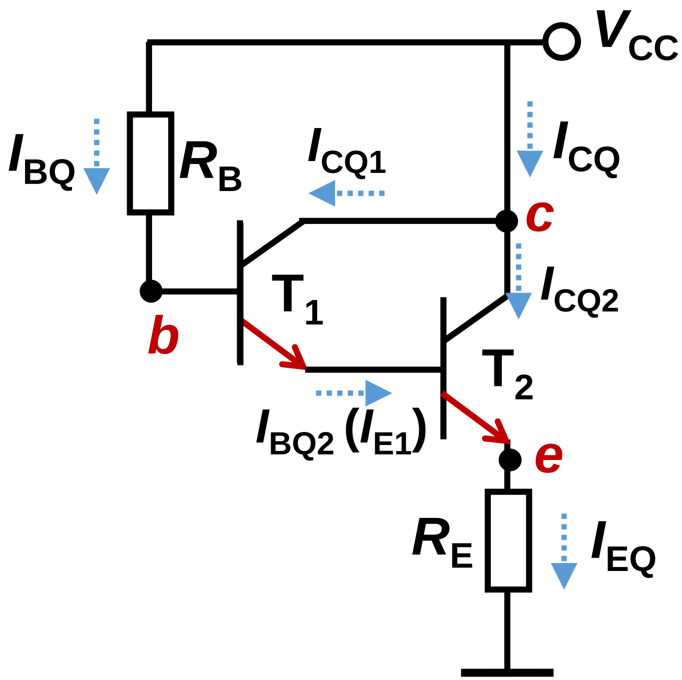
$$\begin{cases} U_{BEQ} = U_{BEQ1} + U_{BEQ2} \text{ is known} \\ I_{CQ} = I_{CQ1} + I_{CQ2} = \beta_1 I_{BQ} + \beta_2 I_{BQ2} \\ I_{BQ2} = I_{E1} = (1+\beta_1) I_{BQ} \end{cases}$$

$$I_{CQ} = I_{CQ1} + I_{CQ2} = \beta_1 I_{BQ} + \beta_2 (1+\beta_1) I_{BQ}$$

$$I_{BQ} = \frac{V_{CC} - U_{BEQ}}{R_B + (1 + \beta_1 + \beta_2 + \beta_1 \beta_2) R_E}$$

I_{CQ} and I_{EQ}

Analyze DC circuits

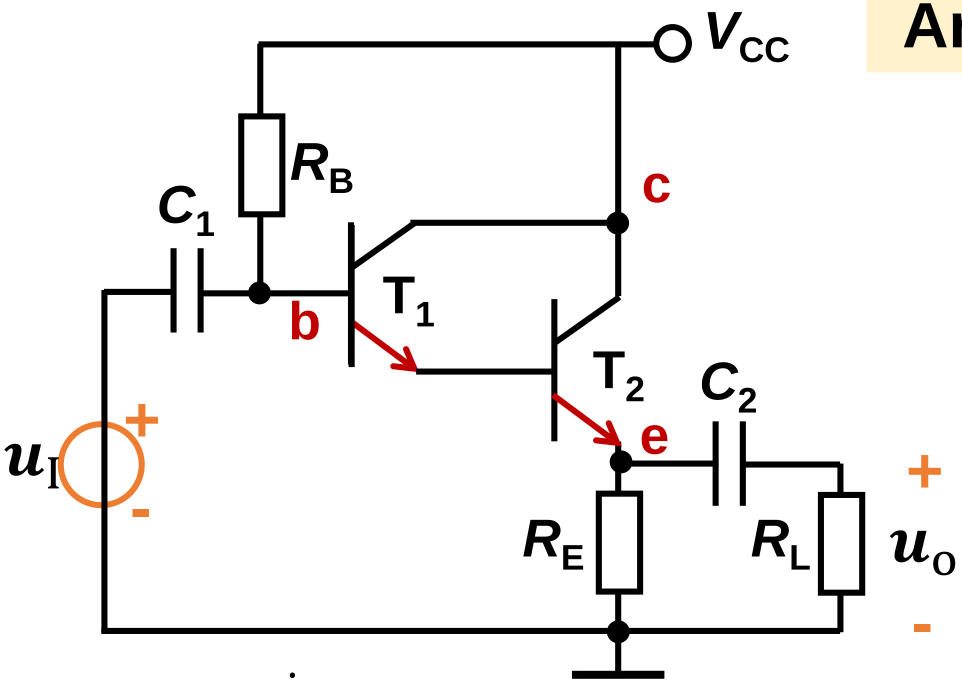


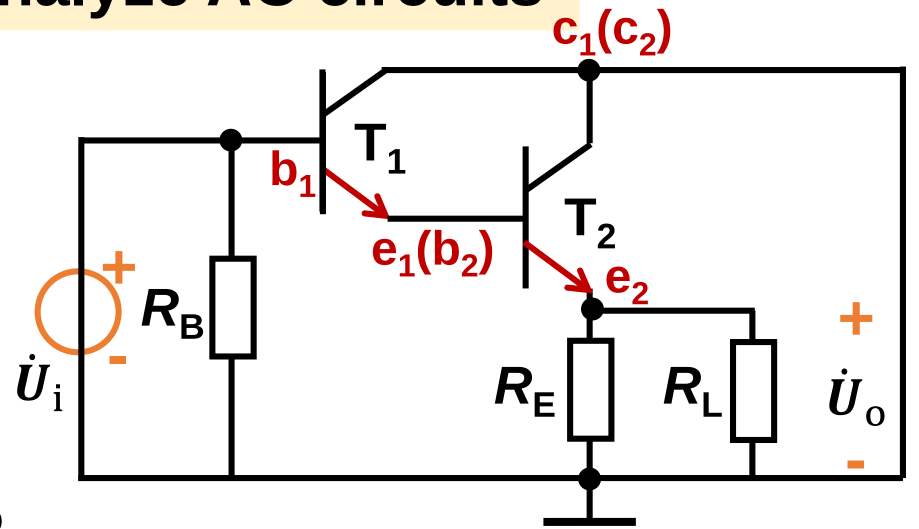
Output circuit equation

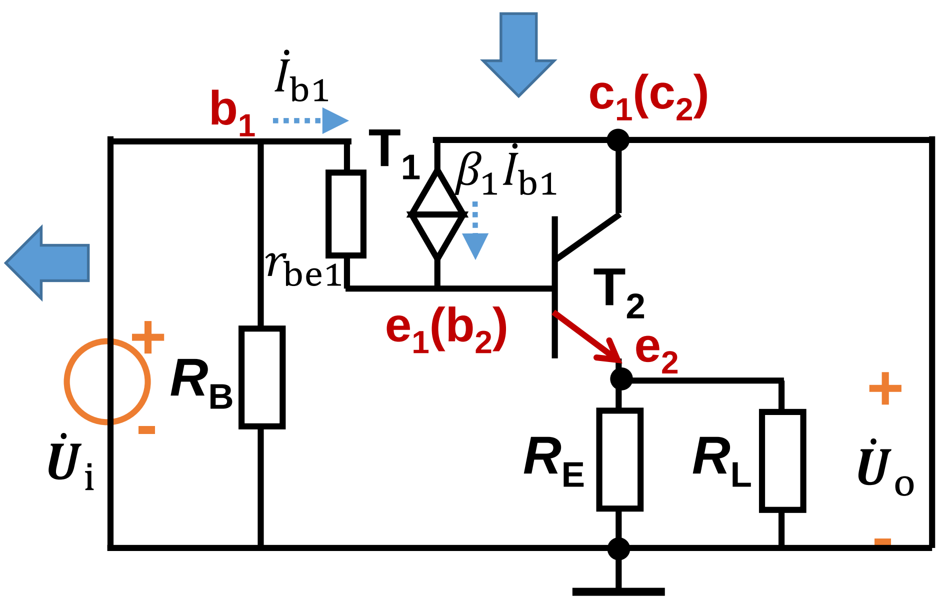
$$V_{CC} = U_{CEQ} + I_{EQ} R_E$$

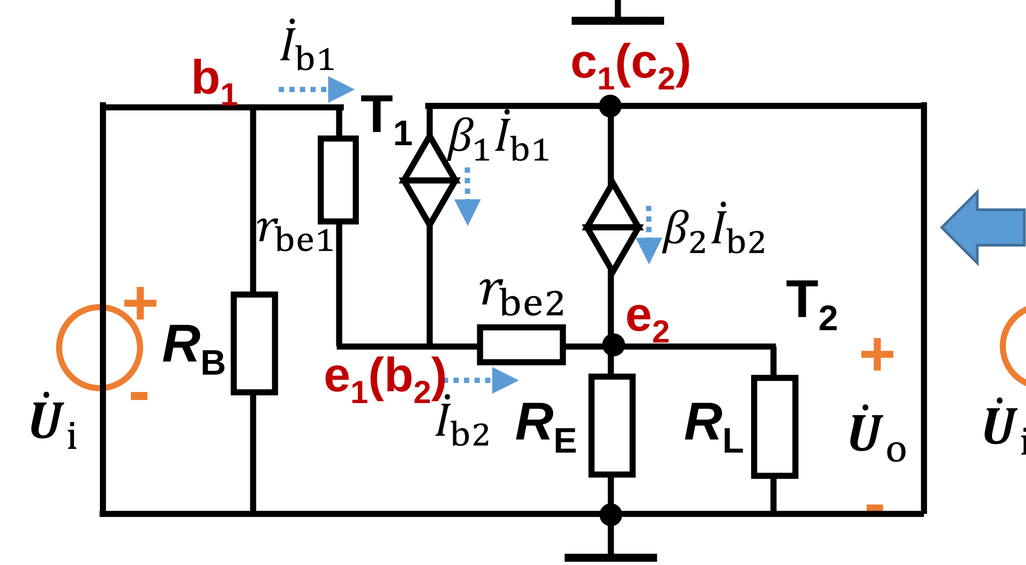
$$U_{CEQ}$$

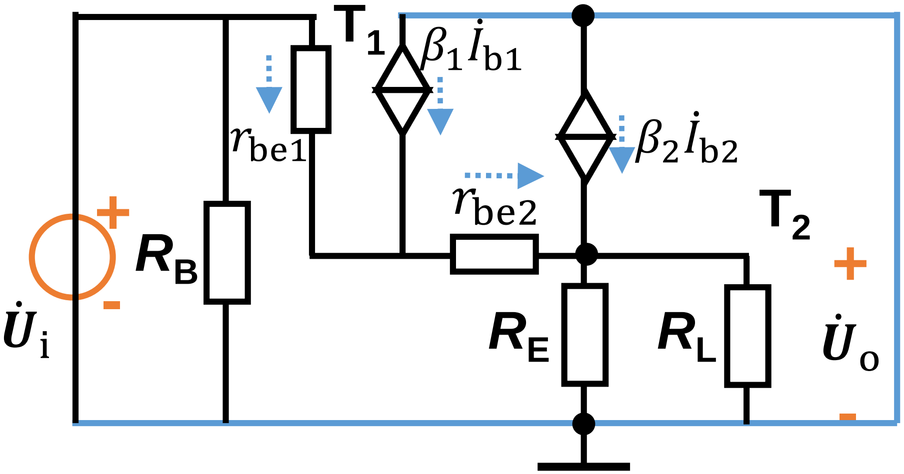
Analyze AC circuits

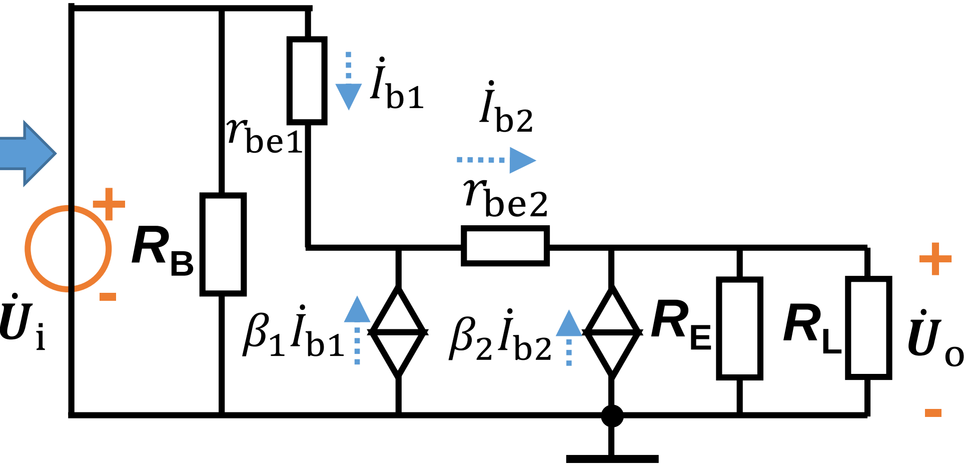


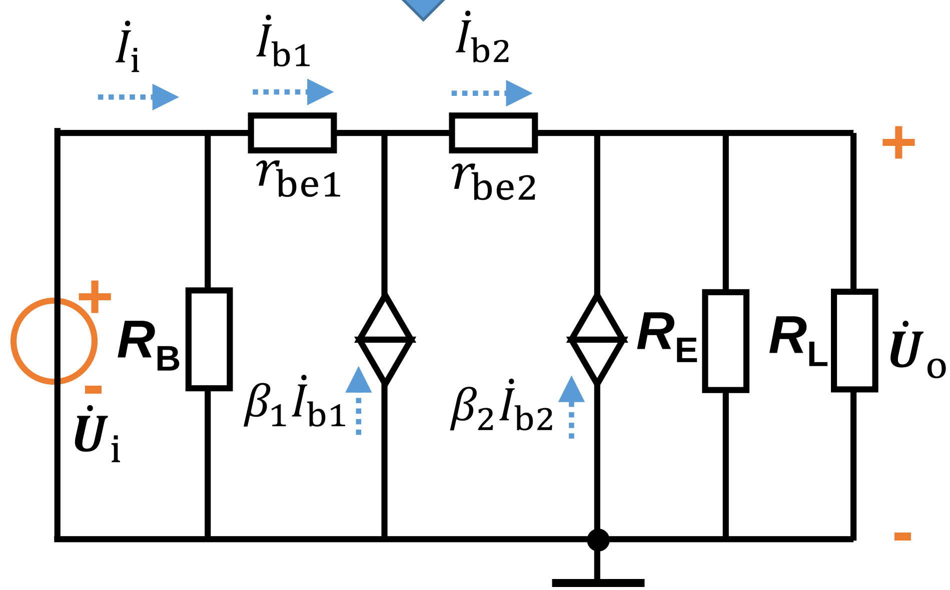












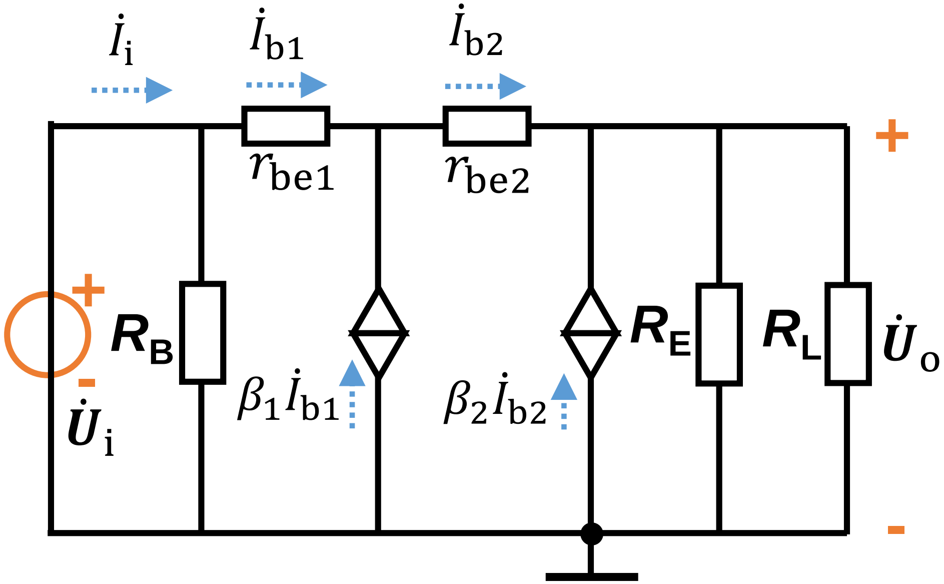
Voltage gain: $\dot{A}_{\mathbf{u}} = \dfrac{\dot{U}_{\mathbf{o}}}{\dot{U}_{\mathrm{i}}}$

$$\dot{I}_{\mathrm{b2}} = (1+\beta_1)\dot{I}_{\mathrm{b1}}$$

$$\dot{U}_{\mathrm{o}} = (1+\beta_2)\dot{I}_{\mathrm{b2}}(R_{\mathrm{E}}||R_{\mathrm{L}})$$
$$= (1+\beta_1)(1+\beta_2)\dot{I}_{\mathrm{b1}}(R_{\mathrm{E}}||R_{\mathrm{L}})$$

$$\dot{U}_{\mathrm{i}} = \dot{I}_{\mathrm{b1}}r_{\mathrm{be1}} + \dot{I}_{\mathrm{b2}}r_{\mathrm{be2}} + \dot{U}_{\mathrm{o}}$$
$$= \dot{I}_{\mathrm{b1}}r_{\mathrm{be1}} + (1+\beta_1)\dot{I}_{\mathrm{b1}}r_{\mathrm{be2}} + \dot{U}_{\mathrm{o}}$$

$$\dot{A}_{\mathbf{u}} = \frac{\dot{U}_{\mathbf{o}}}{\dot{U}_{\mathrm{i}}} = \frac{1}{1+\dfrac{r_{\mathrm{be1}}+(1+\beta_1)\,r_{\mathrm{be2}}}{(1+\beta_1)(1+\beta_2)(R_{\mathrm{E}}||R_{\mathrm{L}})}}$$



Input resistance: $\boldsymbol{R}_{\mathrm{i}} = \dfrac{\dot{\boldsymbol{U}}_{\mathrm{i}}}{\dot{\boldsymbol{I}}_{\mathrm{i}}}$

$$\dot{I}_{b2} = (1+\beta_1)\dot{I}_{b1}$$

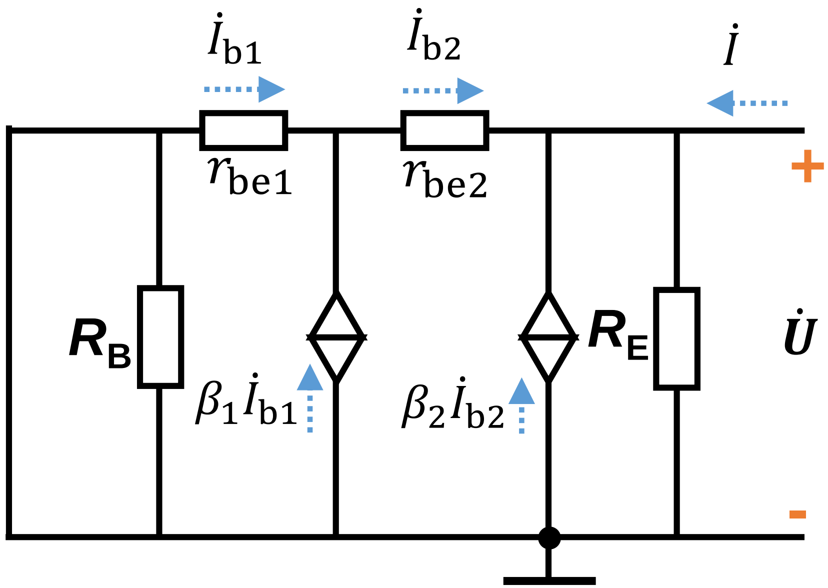
$$\dot{U}_o = (1+\beta_2)\dot{I}_{b2}(R_E||R_L)$$
$$= (1+\beta_1)(1+\beta_2)\dot{I}_{b1}(R_E||R_L)$$

$$\dot{U}_i = \dot{I}_{b1} r_{be1} + \dot{I}_{b2} r_{be2} + \dot{U}_o$$
$$= \dot{I}_{b1} r_{be1} + (1+\beta_1)\dot{I}_{b1} r_{be2} + \dot{U}_o$$

$$\dot{I}_i = \dot{I}_{b1} + \dot{U}_i/R_B$$

$$R_i = \frac{\dot{U}_i}{\dot{I}_i} = \frac{\dot{U}_i}{\dot{I}_{b1} + \dot{U}_i/R_B} = \frac{1}{\dot{I}_{b1}/\dot{U}_i + 1/R_B}$$

$$= R_B \,//\, [r_{be1} + (1+\beta_1) r_{be2} + (1+\beta_1)(1+\beta_2)(R_E||R_L)]$$



Output resistance: $R_o = \dfrac{\dot{U}}{\dot{I}}\Big|_{R_L=\infty,\dot{U}_i=0}$

$$\dot{I}_{b2} = (1+\beta_1)\dot{I}_{b1}$$

$$\dot{U} = -\dot{I}_{b1}r_{be1} - \dot{I}_{b2}r_{be2}$$
$$= -\dot{I}_{b1}r_{be1} - (1+\beta_1)\dot{I}_{b1}r_{be2}$$

$$\dot{I} = -(1+\beta_2)\dot{I}_{b2} + \dot{U}/R_E$$
$$= -(1+\beta_2)(1+\beta_1)\dot{I}_{b1} + \frac{\dot{U}}{R_E}$$

$$R_o = \frac{\dot{U}}{\dot{I}} = \frac{\dot{U}}{-(1+\beta_2)(1+\beta_1)\dot{I}_{b1} + \dot{U}/R_E} = \frac{1}{-(1+\beta_2)(1+\beta_1)\dot{I}_{b1}/\dot{U} + 1/R_E}$$
$$= \frac{1}{\dfrac{(1+\beta_2)(1+\beta_1)}{r_{be1} + (1+\beta_1)\,r_{be2}} + 1/R_E}$$